

Beyond Natural-Language Interfaces: Decisions, Scores, and Predictive Distributions

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Abstract

Language models increasingly supply the predictions on which other systems make decisions. For these systems, an interface defines the predictive information passed to a consumer, yet output format alone does not establish its statistical meaning or computational cost. This survey connects predictive interfaces across classification, reward and quality scoring, numerical regression, and probabilistic forecasting. We organize the literature around the target being reported, the transformations applied to a report, and the computation required to produce it. Drawing on statistical foundations and comparisons between concrete method configurations, we explain why accurate ranking need not provide calibrated probabilities, structural validity need not improve task performance, and compact outputs can conceal extensive generation. We also distinguish uncertainty about observed outcomes from uncertainty in a teacher’s generation process. These distinctions connect interface design to the information a downstream decision actually requires: an appropriate statistic may suffice for a fixed objective, while changing objectives can require richer retained reports or renewed validation. The resulting taxonomy and selection guide link predictive uses to feasible production mechanisms and the evidence needed to compare them. We propose evaluation protocols for changes in losses, constraints, and teacher processes. The unifying principle is to design interfaces around the decisions their reports must support, and to make explicit what must be revalidated when the intended use changes.

Keywords: language models; predictive interfaces; decision rules; scoring; numerical prediction; calibration; evaluation.

1. Prediction beyond conversational answers

A language model can help a system by choosing a category, routing a request, scoring a response, predicting a price, or reporting forecast quantiles. Its consumer may need a usable prediction rather than an explanation. Yet visible format is a poor guide to the resulting system. A vocabulary head can select a single class [1], while a scalar verifier score can require dozens of generated rationales [2]. A model can also evaluate a supplied string’s likelihood without sampling that string [3]. Describing all these mechanisms as generative or non-generative conflates training, execution, and delivery.

We use **predictive interface** for the declared boundary at which a model system supplies a report to a prediction-related consumer, including the transformations that produce that report. A verbalizer, numeric parser, regression head, sample statistic, or constrained solver can therefore belong to the interface. An inspectable internal vector does not automatically constitute a delivered report. Boundaries must be explicit: a router can deliver a model identifier even though the selected model subsequently writes a conversational answer.

Figure 1 connects the delivered reports to illustrative consumer uses and the three design questions developed in this review.

The review develops three practical conclusions, each attached to a controlled example and a boundary.

- **Choose the report for the target and loss.** LM-BFF and RAFT parameterize points; RAIL changes the statistic of fixed samples [4], [5], [6]. xVal’s error/copying contrast and the calibration/CRPS contrast for price quantile regression show why the desired property must be explicit [7], [8].
- **Validate the transformed property.** Existing outputs can support useful changes before replacing a backbone: contextual correction improves label decisions [9], while the fixed-score ILP for structured prediction improves validity with either higher or lower F1 [10]. Grammar-aligned sampling can raise or lower correct-sample counts depending on the problem [11].

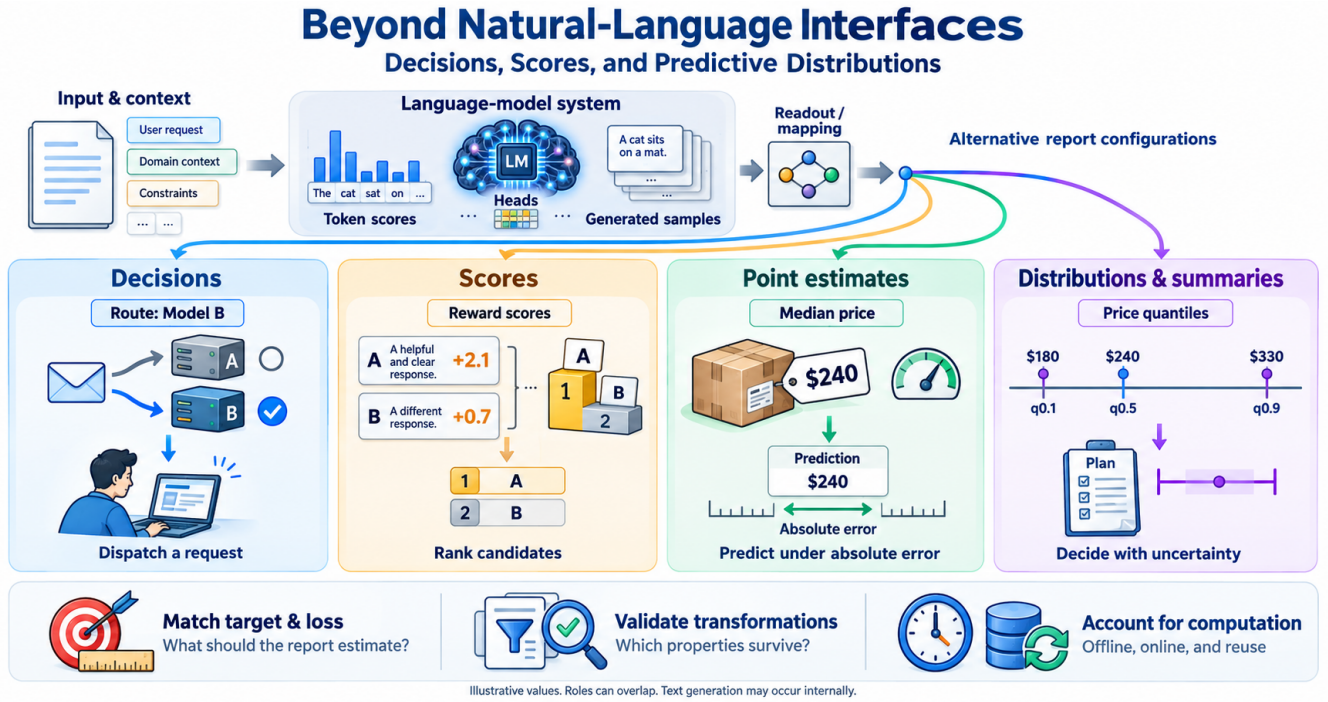


Figure 1. Overview of predictive interfaces. A language-model system delivers a report through a readout or mapping to a downstream consumer. Four parallel, illustrative configurations show a routing decision, reward scores used to rank candidates, a median price prediction evaluated under absolute error, and a finite set of price quantiles used in a decision requiring numerical uncertainty. These roles may overlap. Token scores, trained heads, and generated samples denote possible production mechanisms; each configuration need not use all of them. The lower band summarizes three questions developed in the review: whether the report matches the target and loss, which properties survive its transformations, and where computation is incurred or reused. Reward scores are not identified with correctness probabilities. Finite quantiles do not specify a full density or automatically guarantee coverage. All values and geometry are schematic; quantile-marker spacing and the interval icon are not to scale. Text generation may occur inside the producer even when the delivered report is compact.

- **Account for the execution schedule.** Offline teacher-probe labels, RELISH’s fixed memory, and CAMEL’s conditional reflection provide distinct forms of reuse [12], [13], [14]. Twenty-rating and 32-rationale paths show why a scalar endpoint can retain substantial online work [15], [2].

The reader-facing contributions are a consumer-to-interface guide, a 29-study method matrix, and an inspectable configuration record with source-level provenance. Their value is cross-domain evidence integration. Statistical decision rules and consumer-specific aggregation already have established theoretical and empirical predecessors.

1.1 Scope and exclusions

The core scope comprises finite-label classification, reward and quality scoring, real-valued prediction, and numerical predictive distributions. Language may appear in inputs, intermediate rationales, or output serialization. A digit string is a numerical interface when its consumer interprets it as a number. Point estimates remain explicit because their losses and units distinguish them from ordinal scores. Routing appears as a representative pre-answer decision, not as a comprehensively surveyed application area.

Structured prediction provides a bounded composition case: local scores can form a constrained assignment, and task heads can populate spans or fields [10], [16]. Grammar-based methods expose the difference between valid objects and samples from a model conditioned on validity [11], [17]. These cases clarify interface properties without extending the review to all extraction or program synthesis.

Selected architectural bridges broaden the mechanism comparison. xVal and Chronos train scientific or time-series models rather than merely attaching heads to frozen general-purpose LMs [8], [18]. The classification families in [19] and numerical decoders in [20] likewise have distinct training provenance. Their results inform representation choices, not an isolated estimate of the value of natural-language pretraining. By contrast, text-conditioned quantile regression adapts pretrained LMs to recorded prices [7], while LLM Processes constructs numerical predictions from frozen models [21].

Broad latent reasoning, multi-agent communication, action prediction, and multimodal generation have separate organizing problems and remain outside this review.

1.2 Relationship to prior work

Output spaces and their mappings are established topics. *Pre-train, Prompt, and Predict* studies answer engineering [22]; *The Prompt Report* addresses answer shape, space, extraction, and goal-aware evaluation [23]. Judge, reward, and uncertainty surveys already discuss scores, aggregation, reliability, and computation. The present review inherits these distinctions rather than claiming they were overlooked.

Table 1. Adjacent survey traditions and this review’s organizing emphasis. The differences concern emphasis, not exclusive ownership of a topic.

Literature	Established contribution	Connection developed here
Prompting and answer engineering [22], [23]	Answer spaces, verbalizers, extractors, prompting and learning	Follow mappings into statistical targets, numerical reports, and trained heads
Reward models and judges [24], [25], [26]	Scalar and pairwise evaluation, aggregation, biases	Compare their score semantics with classification and numerical prediction
Confidence and calibration [27], [28]	Uncertainty estimation, calibration, selective prediction	Track the particular event and the effect of transformations on its use
LMs for tabular data [29]	Prediction and generation across applications	Separate numerical encoding, readout, loss, and decision rules

The recent numeracy survey [30] already organizes representation, arithmetic, comparison, and structural tasks alongside architecture, data, training, and inference. Its coverage includes pretrained compatibility and budgets for input, thinking, and output. These axes are therefore established context. Our emphasis is on tracing outcome versus teacher targets, delivered statistical reports, and consumer transformations through specific evaluation configurations.

Individual methods are equally important predecessors. RAIL explicitly chooses numerical statistics for prediction losses [6]. G-EVAL aggregates rating probabilities [15]. TrustJudge connects distribution-sensitive scoring with rating-comparison inconsistencies [31]. Numerical predictive distributions and text-to-distribution learning are already explicit objectives in [32], [7], [21]. Decision theory, proper scoring rules, and property elicitation supply the underlying statistical relationships [33], [34], [35], [36].

Our contribution is a cross-domain evidence synthesis: connect these mechanisms using the same configuration-level questions, identify which measured properties survive a proposed change, and trace the computation required to supply the report. This yields explanations supported by comparisons and counterexamples rather than another catalogue of output forms. Appendix A documents the search and inclusion decisions; Appendix E binds source locators to inspected versions.

The closest consumer-aware methodological predecessor is Xiong et al.’s comparison of uncertainty elicitation and aggregation [37]. It already separates calibration from failure detection and recommends different aggregators. Our guide extends that established principle across decisions, numerical properties, distributional targets, access requirements, and the controls supporting each comparison; it does not claim to originate consumer-dependent evaluation.

1.3 Corpus and review procedure

The evidence comprises **29 configuration-coded method studies**, **four statistical-theory sources**, **eight positioning surveys**, a focused numeracy-survey comparison, and **16 direct predecessors**. The method core supplies 108 representative configurations, 53 empirical comparisons, and two diagnostics. Predecessors support the specific mechanisms and attributions cited; they do not enlarge the configuration count.

This is a question-driven synthesis using targeted searches and citation following. Appendix A gives the search stages, inclusion criteria and query log; Appendix B compares the core methods. Records distinguish reported findings from deductions and unresolved fields, and preserve the target, fixed and varied components, metric, denominator and budget. Appendix C gives a complete example. Appendix E binds claims to inspected PDF versions separately from publication metadata.

2. Reports, consumers, and statistical meaning

2.1 Specify the delivery boundary

Let Y denote an outcome or supervised target, r the report delivered by the model system, and $a = \delta(r, c)$ the action taken by a consumer with requirements c . An answer-conditioned correctness score, a question-only routing decision, and a vector of forecast quantiles differ in both their information and their intended use. The boundary determines which transformations belong to the producer and which belong to the consumer.

We use D for decisions, S for scores, V for numerical point or property reports, and P for distributions or finite summaries. These roles overlap: one system can expose quantiles, deliver their median, and apply a decision threshold. C records bounded composition into spans, fields, or constrained assignments. Figure 2 maps the configuration-coded core, and Figure B1 in Appendix B records study by study which roles each study realizes and how its coded configurations were fitted; the selection guide in Section 7 states the information needed for each consumer, including coverage-oriented sets as a boundary case.

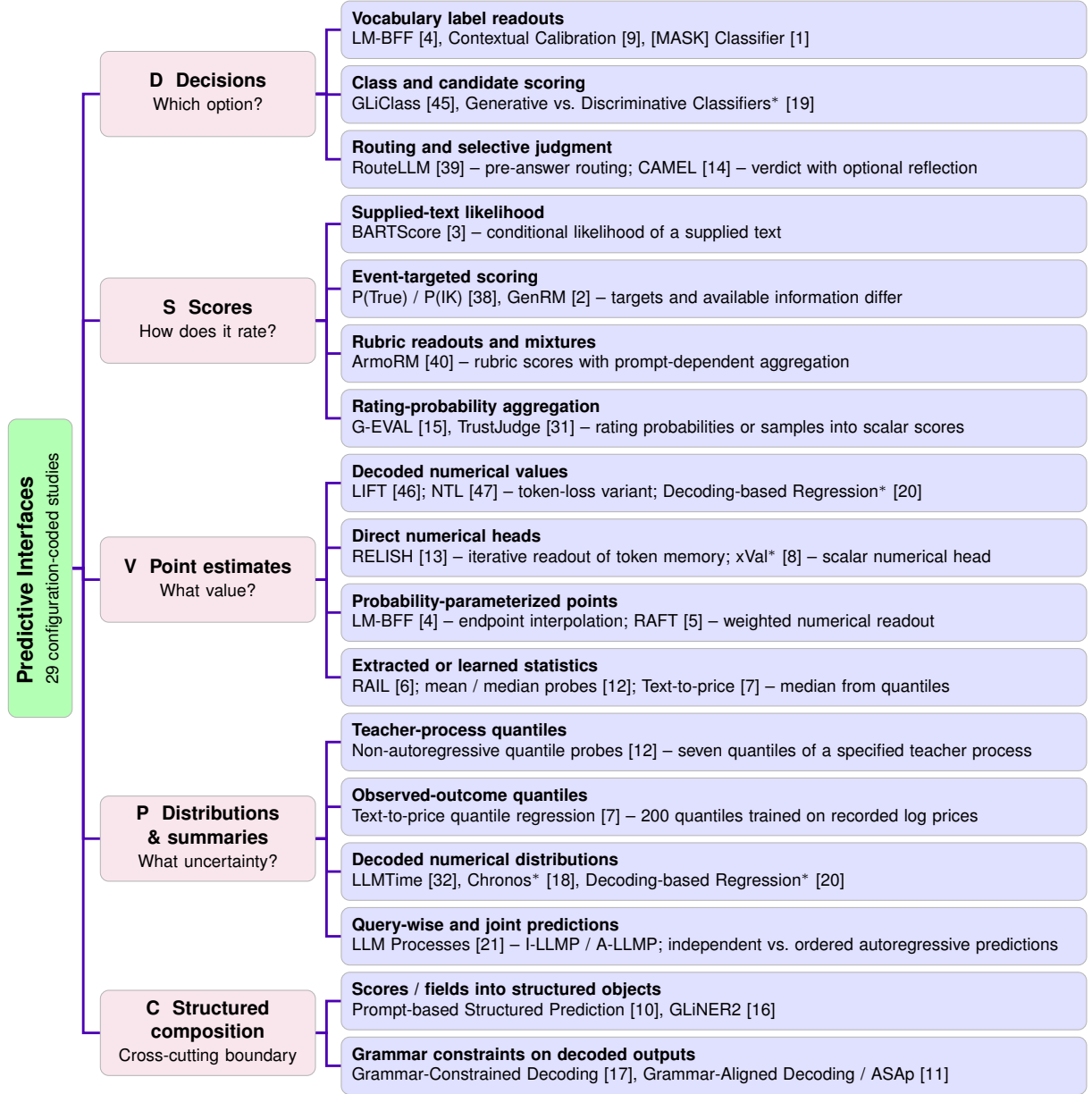


Figure 2. Taxonomy of predictive interfaces. Configuration-level taxonomy of the 29 configuration-coded method studies. The first level organizes the delivered report: decisions (D), scores (S), point estimates (V), and distributions or finite summaries (P); structured composition (C) is drawn as a cross-cutting boundary rather than a fifth disjoint role. Each leaf is one of the 17 mechanisms by which a report is produced, headed by its name, and lists the studies whose selected configurations realize it. Roles overlap, so a study can appear under two branches (LM-BFF under D and V; [12], [7] and Decoding-based Regression under V and P); Figure B1 in Appendix B shows this overlap study by study together with the training provenance of the coded configurations. [12] learns teacher-process quantiles; [7] learns quantiles from recorded outcomes. Finite quantile sets do not specify a full density. Asterisks mark architectural bridges: xVal, Chronos and Decoding-based Regression train from scratch; [19] compares separately trained classifier families and includes pretrained-model analyses. The foundational verbalized-confidence studies discussed in the text are adjacent predecessors and are not configuration-coded here. Edges indicate grouping, not execution or ranking. Text generation may still occur internally. Bracketed numbers identify references in the survey bibliography.

2.2 Let the objective identify the report

A report may estimate a property $T(P)$ rather than an entire distribution. Under appropriate moment and minimizer conditions, squared error elicits a mean, absolute error admits medians, and pinball loss elicits its specified quantile [33]. A Huber-trained head should be described by its objective, as in RELISH [13]. With a suitable distribution q , a consumer may minimize $\mathbb{E}_q[L_c(a, Y)]$; direct predictors learn the required report without implementing this intermediate distribution.

A proper distribution loss gives the target distribution an optimal *population* expected score, uniquely for strict propriety on the stated class [34]. Empirical calibration remains a property to evaluate after estimation. Composite-loss theory specifies the link between internal scores and probabilities [35]; a sigmoid on an arbitrary margin supplies a numerical range, not the event semantics or validation. Vector-valued elicitation supports several properties jointly [36]. A finite quantile vector supports its specified decisions while leaving interpolation and tails to an additional model.

The design rule is to train and evaluate the property the consumer uses. LM-BFF’s endpoint probabilities parameterize a similarity estimate; RAFT places numerical error on a probability-weighted scalar [4], [5]. Their point objectives differ from learning a full outcome law. Appendix B records target, numerical intervention, readout, access, and controls for each core study.

2.3 Name the probability’s event and population

The probability of a rating token, answer correctness, and a teacher emitting a number refer to different random variables. $P(\text{True})$ conditions on a supplied answer, whereas $P(\text{IK})$ predicts success of a specified answering process from the question [38]. Pre-answer routing and completed-answer verification consequently have different information [39], [2]. ArmoRM’s normalized gate weights play another role: they mix rubric predictions [40].

Verbalized confidence is a direct predecessor to this analysis. A generated percentage can be trained to express uncertainty, elicited from an aligned model, or aggregated across samples [41], [42], [37]. These studies evaluate how the report predicts correctness. Xiong et al. explicitly distinguish probability calibration from failure detection and give different aggregation recommendations for these consumers [37]. Consumer-dependent evaluation is therefore inherited; our synthesis connects it to decisions, numerical reports, access, and comparison controls across domains. Semantic uncertainty similarly aggregates probability over equivalent answers before computing uncertainty [43], [44]; Section 5 develops that mapping.

Target provenance is consequential. The teacher-probe study [12] learns properties of a specified teacher generation process; the price-quantile study [7] learns quantiles from recorded prices. LLMTIME and Chronos evaluate forecasts against subsequent observations [32], [18]. Coverage and accuracy belong to those particular populations. Joint reports also need their dependence structure specified: LLM Processes distinguishes independent marginals from an ordered autoregressive construction [21].

2.4 Keep claims attached to configurations

Configurations distinguish reports with different targets or execution paths; comparisons state which components are held fixed. Appendix C demonstrates how a real record preserves epistemic status and source-level evidence, while Appendix E identifies the inspected versions.

3. Producing decisions and scores with language models

This section compares vocabulary readouts, supplied-string scoring, rating aggregation, and fitted heads. The useful distinction is what each path observes, learns, and delivers. Figure 3 makes the supervision target explicit alongside the execution path.

3.1 Vocabulary readouts and candidate-conditioned representations

LM-BFF normalizes masked-position logits over selected label words and maps those words to classes [4]. The denominator is the candidate vocabulary, rather than the entire language vocabulary or all synonymous strings. For STS-B, two endpoint words instead parameterize a similarity estimate: their weights are trained from the continuous target and interpolated into a point prediction [4]. These weights can be uniquely determined by that two-endpoint representation without being a posterior that real similarity takes only the endpoint values. Probability parameterization and outcome-distribution estimation are distinct questions.

Single-position instruction classification extends the vocabulary route. *It’s All in The [MASK]* retains ModernBERT’s masked-language-model head and trains one answer token, using semantic verbalizers or candidate identifiers [1]. The training set is filtered to fit this interface. Its MMLU ablation changes performance substantially while retaining the readout and changing the training-objective mixture [1]. The complete recipe, rather than output length alone, supports the reported capability.

GLiClass contextualizes input and candidate descriptions jointly, then scores pooled text and label representations [45]. Labels can change without assigning each class a permanent vocabulary token. Candidate sets also enter representation computation: this differs from scoring a fixed set of independently encoded labels. The model’s multi-stage training and candidate-dependent work belong to the configuration; a dot-product endpoint does not imply constant computation as the label set grows.

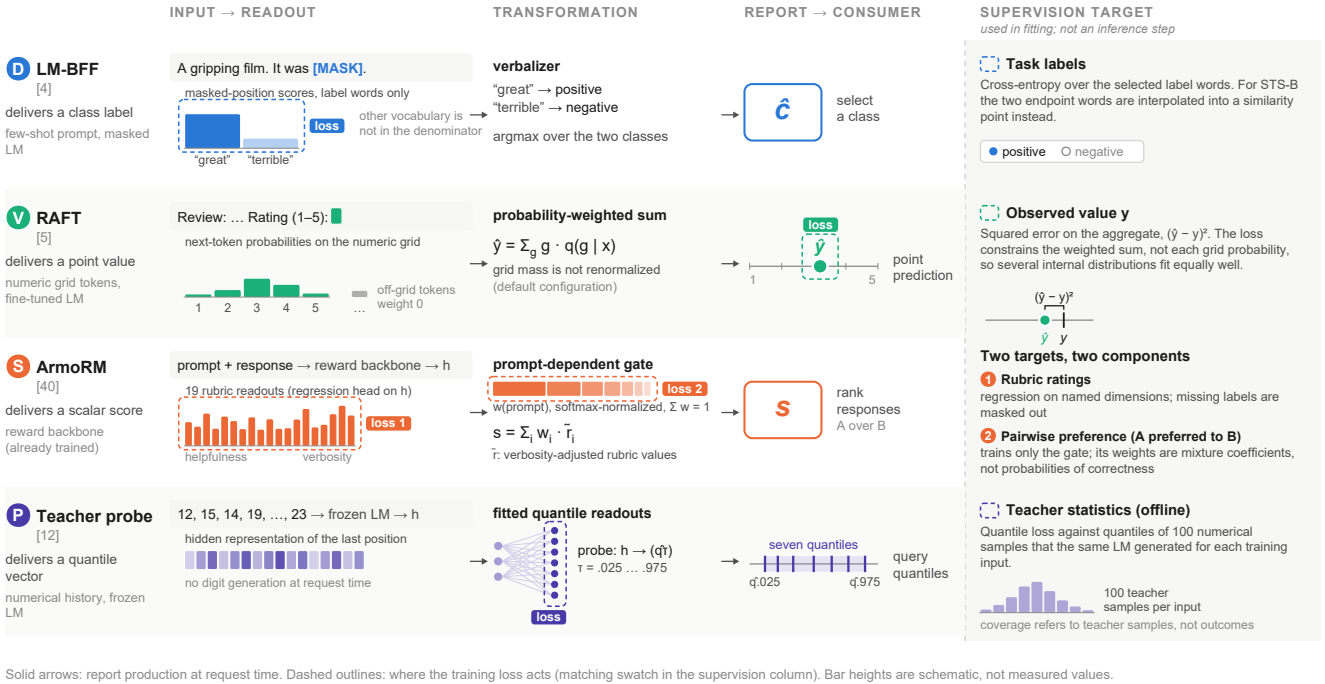


Figure 3. The training target gives a report its meaning. Four configurations read left to right as request-time production paths (input and readout, transformation, delivered report and its consumer); the shaded right-hand column shows the target used in fitting, which is not an inference step. Dashed outlines mark where the training loss acts, with a matching switch beside each target. LM-BFF [4] normalizes masked-position scores over the label words only and maps them to a class with a verbalizer; the cross-entropy acts on those selected-token scores. RAFT [5] computes a probability-weighted sum over the numeric grid without renormalizing the grid mass; the squared error acts on the aggregate, so the loss constrains the weighted sum rather than each grid probability. ArmoRM [40] reads nineteen rubric values from an already trained reward backbone and mixes verbosity-adjusted values with a prompt-dependent, softmax-normalized gate; rubric ratings train the readouts (loss 1) and pairwise preferences train only the gate (loss 2), whose weights are mixture coefficients, not correctness probabilities. The teacher probe [12] maps the frozen model’s hidden representation to seven quantiles ($\tau = 0.025$ to 0.975) fitted against statistics of 100 teacher samples generated offline for each training input; its coverage refers to teacher samples, not outcomes. Bar heights and positions are schematic, not measured values.

Table 2. Representative production routes. Rows describe operations; one system can combine them or expose more than one report.

Route	Information supplied	Operation and delivered report	Examples
Candidate vocabulary readout	Input, template, candidate words	Selected-token probabilities to category or point value	LM-BFF; single-mask classification [4], [1]
Candidate-conditioned features	Input and label descriptions	Representation scores to one or several labels	GLiClass [45]
Known-string scoring	Conditioning text and supplied target text	Token log probabilities to quality score	BARTScore [3]
Rating elicitation	Rubric, candidate and evaluation instructions	Rating probabilities or samples to scalar summary	G-EVAL [15]
Correctness verification	Question and completed answer	Answer-label probability to candidate score	P(True); GenRM [38], [2]
Learned numerical readout	Question alone, or question plus answer	Success estimate or named rubric values	P(IK); ArmoRM [38], [40]

These routes have no universal performance ordering. Under LM-BFF’s few-shot protocol, manual prompting improves STS-B but loses to ordinary fine-tuning on CoLA: 9.3 (standard deviation 7.3) versus 33.9 (14.3) Matthews correlation on the reported percent scale [4]. Conversely, a substantial improvement can occur without updating the backbone. Contextual calibration estimates prompt-specific label biases from content-free inputs; for GPT-3 175B, four-shot AG News accuracy changes from $61.0 \pm 10.9\%$ to $85.9 \pm 1.3\%$ [9]. The $\pm$ values are standard deviations across five randomly selected training-example sets at a fixed prompt format; the same sets are used before and after label-bias correction.

3.2 Scoring a supplied string

BARTScore evaluates the conditional log probabilities of text already available to the evaluator [3]. It need not generate an assessment sentence or sample the target text. Source-to-hypothesis scoring measures a different conditional event from reference-to-hypothesis or hypothesis-to-reference scoring. The paper names the latter directions precision and recall, but they

are likelihood-based analogues rather than counts of correct items. Direction, length handling and aggregation determine the score’s interpretation.

The prompt ensemble averages target-token log probabilities within each prompt and then averages over prompts [3]. Its output is therefore not a full-string probability. Fine-tuning and prompt choice change its relationship with human quality: further paraphrase training improves the reported WMT19 correlation but reduces Rank19 factuality accuracy relative to the CNN/DailyMail model [3]. A probability-based origin supplies a scoring mechanism; usefulness for another quality criterion remains empirical.

Generative classification can similarly evaluate the supplied input under candidate classes and select a label without generating that input at prediction time [19]. This differs from generating a label string autoregressively. The controlled families in that study largely train from scratch and use different training schedules, so they illustrate mechanisms without isolating a frozen LLM’s output-head effect. The operational distinction is whether the sequence being scored is known, not whether the underlying model admits an autoregressive factorization.

3.3 Rating distributions and the resolution of scores

G-EVAL generates task-level evaluation steps, incorporates them into a rating prompt, and aggregates possible ratings [15]. Its text-davinci-003 configuration reads token probabilities. The inspected GPT-4 configuration instead estimates rating frequencies from twenty samples at temperature 1 and top-p 1 because the required probabilities were unavailable through that API. A frequency-weighted mean of valid ratings is the sample mean; these implementations deliver similar scalars through different access and computation paths.

The finer scalar changes evaluation resolution. On SummEval, GPT-4 probability aggregation changes mean Spearman correlation from 0.502 to 0.514, whereas Kendall correlation changes from 0.446 to 0.418 [15]. The table caption attributes the apparently favorable Kendall score without aggregation to many ties and cautions against treating it as a fair assessment of evaluation ability. This is evidence that the measurement protocol and score resolution interact, rather than a clean demonstration that two real consumers prefer opposite methods. The xVal lookup comparison in Section 4 provides a more direct contrast between approximate and precise numerical requirements.

BARTScore and G-EVAL also expose different probability referents. One concerns candidate wording under a conditional generation model; the other concerns ratings emitted under an evaluation prompt. Correlation with human judgments validates aspects of their delivered scoring procedures. It does not identify a human-disagreement distribution or establish a common cardinal scale across rubrics.

3.4 Supervised correctness and multidimensional reward

P(True) receives a question and completed answer before reporting a True-label probability; its multi-sample configuration additionally supplies other sampled answers [38]. GenRM trains an analogous vocabulary readout using Yes/No correctness targets, while its discriminative comparator learns a scalar sigmoid with binary cross-entropy [2]. Both can supply a candidate selector, but their parameterizations and training objectives differ.

GenRM-CoT adds verification-rationale supervision and averages Yes probabilities after sampled rationales, using 32 rationales per candidate by default [2]. This is soft-score aggregation rather than hard voting. Direct GenRM is comparable to or slightly better than the discriminative comparator in the reported additional evaluation; larger gains involve the rationale configuration [2]. Teacher reference answers help construct mathematical verification rationales during training and are absent at deployment. The useful scalar can consequently conceal substantial language computation and privileged training information.

P(IK) acts earlier. It predicts the success of a specified answering process from the question before a particular answer is supplied. Thirty sampled answers per training question estimate that process’s success, and the language model and a separate value head are trained jointly [38]. The trainable backbone and question-only input distinguish this configuration from a frozen probe and an answer-conditioned verifier. Its target is the specified answering process’s success.

ArmoRM separates named evaluation dimensions from their final combination [40]. A regression head produces nineteen rubric values from an answer-conditioned hidden state; a question-conditioned gate combines adjusted values into a scalar reward. The added components are fitted on an already trained reward backbone. Missing rubric labels are masked from regression training, and similar dimensions can retain different annotation sources [40]. Rubric values, normalized mixing weights and the preference probability used to train aggregation are three distinct objects. Its fixed-predictor aggregation ablation, summarized in Section 6, supports this decomposition empirically without turning the gate into a correctness probability.

3.5 Bounded structures as composed reports

Local predictive reports can also populate a bounded object. [10] combines pairwise or label scores using constrained inference; GLiNER2 predicts schema-conditioned spans, classes and fields, then constructs a serialized record in software [10], [16]. The latter includes hierarchical parent counts and occurrence-specific fields, but component NER and classification results do not establish complete hierarchical-record accuracy.

Grammar-constrained decoding restricts a token path, sometimes using input-dependent candidate information [17]. Supplying mention-specific entity candidates can change the information available, alongside enforcing syntax. Grammar-Aligned Decoding additionally distinguishes local reachability of valid completions from preserving the original model’s complete-string distribution conditional on validity [11]. Composition also makes the software consumer concrete: downstream code receives a complete assignment or record, so schema checks, failed fields, and solver outcomes belong in evaluation alongside local prediction accuracy. Section 5 distinguishes structural legality, sampling-law fidelity, and semantic success. Coverage-oriented sets in Section 7 add a different consumer requirement, with their own calibration assumptions; they do not expand this review into unrestricted structured generation.

4. Numerical reports: geometry, targets, and decisions

Numerical prediction separates implementation from meaning especially clearly. A scalar can be decoded from text, read directly from hidden states, or computed from a vocabulary distribution. A digit sequence can instead specify numerical uncertainty. The useful organizing axes are **where numerical structure enters**, **which target is learned**, and **what the consumer requires**. These axes avoid treating continuous output as the endpoint of a progression away from tokens.

4.1 Where numerical structure enters

LIFT serializes features and targets, fine-tunes a language model with token cross-entropy, and parses the generated numerical answer [46]. Its regression procedure includes changed-temperature retries and a training-label-mean fallback after persistent parsing failure. The delivered number therefore belongs to the whole procedure. Its task-dependent results establish that text-mediated regression can be useful without an explicitly distance-aware loss.

Subsequent methods introduce numerical structure at different locations. Some change the consumer’s functional; others change the readout, objective or representation. These choices can coexist.

Table 3. Where numerical structure enters selected configurations. The rows identify interventions, not mutually exclusive families or an accuracy ranking.

Method	Mechanism carrying numerical structure	Delivered report	Essential qualification
RAIL [6]	Inference-time functional of samples or candidate mass	Selected point statistic	Functional depends on the prediction loss
RAFT [5]	Numerical vocabulary readout plus aggregate regression loss	Weighted scalar	Default grid mass is not renormalized
NTL [47]	Numerical token values in the training objective	Parsed numerical generation	Token distance differs from whole-value distance
RELISH [13]	Iterative hidden-state readout, Huber objective	Point estimate	Iteration reuses fixed token representations
Quantile price regression [7]	Multi-output head, smoothed quantile loss	200 quantiles and median	Observed log-prices; LoRA; printed loss definition is ambiguous
LLMTime / LLM Processes [32], [21]	Numerical string scoring and decoding	Density, samples, summaries	Scoring and sampling laws need separate specification
Decoding-based Regression [20]	Finite-precision digit factorization	Binned law or decoded point	Regression models trained for the task
xVal / Chronos [8], [18]	Continuous encoding or scaling and quantization	Numerical predictions or forecast summaries	Main scientific models train from scratch

Appendix F contains the optional numerical mechanism atlas; Table 3 is the primary in-text comparison.

Text-mediated regression also includes two direct predecessors with different adaptation requirements. *From Words to Numbers* predicts from in-context input-output examples using pretrained models without gradient updates [48]. OmniPred instead trains a 200M T5 model from scratch across heterogeneous tasks, uses structured sign/digit/exponent tokens, and takes the median of decoded samples [49]. They fill distinct points in the mechanism comparison: text-only access to a pretrained predictor versus a trained numerical language-model architecture. Their evidence does not isolate an output-head change.

4.2 Distribution-aware inference and point-targeted training

RAIL starts from a classical distinction: the most likely numerical answer need not minimize the consumer’s loss [6], [33]. Squared error calls for a mean; absolute error admits a median. A language-model procedure must both approximate a useful distribution and compute the intended functional of it. These are separate sources of error.

A particularly informative contrast fixes sixteen temperature-1 samples. For PaLM-2-S on STS-B, selecting the highest-probability sample gives RMSE 1.019, whereas the ordinary sample mean gives 0.641 [6]. The main table’s weighted mean additionally changes effective temperature; it is a different intervention. Enumeration also has limits: for FLAN-T5-XL, greedy, enumerated-mean and sampled-mean RMSE are 0.713, 0.741 and 0.676 [6]. Neither a larger candidate set nor a distributional interface by itself ensures a better point decision.

RAFT moves numerical aggregation into training [5]. For a numerical grid G , the distinction between RAIL’s candidate-normalized enumeration and default fixed-weight RAFT is

$$\hat{y}_{\text{RAIL,grid}} = \frac{\sum_{g \in G} g q(g | x)}{\sum_{g \in G} q(g | x)}, \quad \hat{y}_{\text{RAFT}} = \sum_{g \in G} g q_{\theta}(g | x).$$

Off-grid vocabulary mass receives weight zero in default RAFT. Its squared loss acts **on the aggregate**, rather than scoring every numerical outcome under the internal distribution. Multiple internal distributions can support the same aggregate. Good point RMSE consequently need not identify a full outcome distribution. The normalized and unnormalized variants have similar STS-B RMSE at the displayed precision [5], which does not make their probability interpretations equivalent.

The direct-head baseline remains competitive. For Gemma-2 2B on Amazon Wireless, RAFT reports RMSE 0.47, a regression head 0.51 and CE with greedy decoding 0.70; on STS-B, RAFT and the regression head both report 0.54 [5]. Parameterization, training objective and decision rule all contribute. A default single-position numerical grid can also yield its scalar without repeated autoregressive generations.

4.3 Distance-aware losses and direct readouts

NTL retains generation but supplements cross-entropy with a numerical-token objective [47]. Its loss supplies distinctions omitted by treating incorrect tokens identically. At the inspected experimental-repository commit, logits are restricted to numerical tokens before softmax; this resolves the literal pseudocode’s normalization ambiguity for that implementation, without identifying every historical training run. Appendix E records the inspected code version alongside the conflicting printed description.

Numerical structure still acts at token positions. On T5-small/rJokes, ordinary-tokenizer RMSE changes from 2.01 with CE to 1.97 with NTL; single-digit tokenization changes the corresponding comparison from 2.02 to 1.81. A scalar MSE head gives 1.82 [47]. Thus the stronger outcome belongs to an encoding-and-loss configuration, while within-tokenizer contrasts isolate a narrower effect. Positional digit errors do not automatically account for place value or carries, and exact-arithmetic results include reversals.

RELISH instead reads a frozen language model’s token representations with a compact iterative head [13]. A query state repeatedly attends to fixed memory, then produces a scalar; the backbone runs once. Its representations are taken before the language output head, not necessarily from the second-to-last Transformer block. Matched-loss, matched-normalization and approximately matched-head-size comparisons support readout improvements, while low-label regimes sometimes favor RAFT. The Huber objective defines a point target; neither internal iteration nor a scalar’s real-valued range supplies an uncertainty interpretation.

4.4 Numerical distributions need a target and a construction

Digit factorization can provide a legitimate numerical distribution. A base- B , K -digit decoder represents B^K bins through K categorical conditionals. Decoding-based Regression connects this construction to histogram estimation under explicit approximation and learning assumptions [20]. Fixed precision trades representational resolution against estimation difficulty; it does not settle the point functional obtained by greedy decoding.

LLMTime applies numerical encoding to frozen pretrained language models [32]. At fixed precision, scored strings can define a piecewise-uniform density using bin width and the scaling Jacobian. The inspected author code also shows that generation and likelihood are processed differently: the LLaMA NLL path does not apply the sampling temperature or nucleus truncation, while the GPT scoring path estimates excluded mass using a limited returned token list. Density, sampled trajectories and final median therefore refer to specified processing paths. We retain this distinction and exclude a precise aggregate Monash improvement whose historical normalization chain remains incompletely traced.

LLM Processes extends the construction to arbitrary query locations and textual side information [21]. I-LLMP multiplies independently prompted marginals. A-LLMP feeds previous queried locations and generated values into later prompts, allowing dependencies but introducing query-order dependence. A fixed-order joint law is distinct from a consistent stochastic process over arbitrary target sets; the paper explicitly withholds that guarantee for A-LLMP. Its terminal-token likelihood construction also does not guarantee equality with numerical-bin mass under its sampling procedure [21]. These are useful probabilistic interfaces whose coherence and sampling semantics remain part of the specification.

Direct quantile learning offers a complementary route. Price regression adapts a pretrained LM with LoRA and a 200-quantile head, using observed log-prices and smoothed pinball loss [7]. The paper’s printed smoothing definition contains an inconsistent coefficient; the implemented objective has not been independently verified, so we use the reported setup and results rather than derive guarantees from that formula. Its median improves Amazon MAPE from the point-head baseline’s 20.81% to 16.86% [7]. This changes the objective and report together: a squared-loss predictor in log space does not automatically recover the conditional mean in price space. The data also undergo LLM-based price filtering in all splits, so the empirical target is that selected population.

The distribution metrics supply a useful counterweight to a single winning score. On Amazon, Mistral’s quantile head has calibration error 0.042 and CRPSS 0.75; the embedding+kNN baseline has lower calibration error 0.010 but lower CRPSS 0.34 [7]. These are different trained pipelines, not an isolated head effect. The comparison shows why marginal quantile coverage and overall distributional scoring remain separate assessments. Finite quantiles also require a stated interpolation and tail policy for arbitrary distribution queries. The appendix proposes monotone increments and interpolation, but does not unambiguously establish their use in the main results.

Teacher-statistic probes learn a different target. [12] builds training labels from 100 numerical generations per input and predicts a mean, median or seven selected quantiles from hidden states. Its nominal 95% interval has approximately 95.4–96.1% coverage of teacher samples in the reported synthetic settings [12]. Real-next-value point error is tested separately: one mean-probe comparison reports MSE 0.0562 against 0.0555 for the sampled teacher mean [12]. The two evaluations answer different questions.

Table 4. Numerical uncertainty reports with different referents.

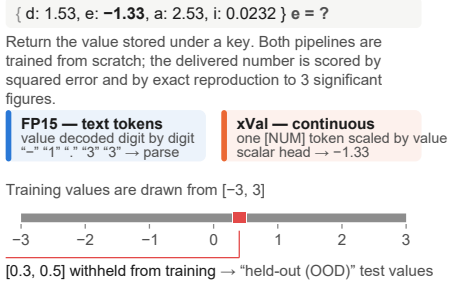
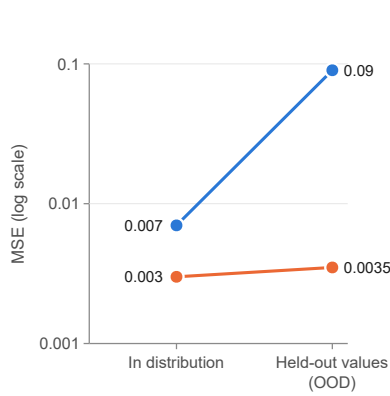
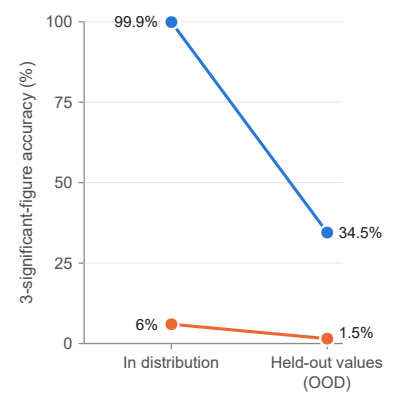
Configuration	Supervision or conditioning	Report boundary	Additional condition for reuse
LLMTime [32]	Pretraining plus observed history in a prompt	Scored density or sampled paths	Numerical support, scaling and scoring/sampling law
I-/A-LLMP [21]	Observed pairs plus optional descriptive text	Independent marginals or ordered joint predictions	Dependence structure and query-order consistency
Price quantile head [7]	Filtered observed log-price labels	200 selected quantiles; median	Monotonicity, interpolation, tails and selected population
Teacher-statistic probe [12]	Statistics of specified teacher generations	Point property or seven quantiles	Teacher process and summaries retained; world-target validation is separate
Chronos [18]	Observed and synthetic time series	Quantized future paths and summaries	Scaling, finite support and forecasting protocol

Forecasting accuracy alone also leaves the contribution of language pretraining unresolved. Tan et al. remove or replace the LLM component in OneFitsAll/GPT4TS, Time-LLM, and CALF while retaining the remaining trainable forecasting pipeline; simpler ablations often match or improve the evaluated results [50]. This supports testing the backbone’s contribution. It is not a direct ablation of LLMTime or scratch-trained Chronos, and should not be presented as refuting those particular report constructions.

4.5 Support and precise copying expose different failure modes

Chronos trains its main time-series Transformers from scratch, scaling by historical mean absolute value and using 4,094 numerical bins [18]. Its representable range is approximately $[-15s, 15s]$, where s is the scale. In the paper’s periodic construction, one unit spike every n observations gives $s = 1/n$. For $n > 15$, a future unit spike lies outside output support [18]. Better token selection cannot recover an unavailable value.

xVal uses continuous numerical input embeddings and a scalar output stream, also training its main scientific models from scratch [8]. Its number-lookup result exposes another failure mode: MSE favors xVal over FP15, 0.003 versus 0.007, while three-significant-figure accuracy favors FP15, 99.9% versus 6% [8]. The same evaluated predictions receive opposite rankings under approximate deviation and precise copying. This does not isolate an output-head effect because the pipelines differ; it shows why the consumer’s numerical requirement must be explicit.

a Number look-up task**b Numerical error****c Exact copying**

Lower MSE favours xVal; exact copying favours FP15.

Same evaluated predictions, two metrics. Encoding, head and training change together, so this is a pipeline comparison, not an isolated output-head effect. Central reported values from [8], Tables 12–13.

Figure 4. Choose the interface for the error that matters. **a**, The number look-up task of [8]: return the value stored under a key. FP15 decodes the value digit by digit as text tokens; xVal represents it with one [NUM] token scaled by the value and a scalar head. Both pipelines are trained from scratch on values drawn from [-3, 3], with [0.3, 0.5] withheld to form the held-out (OOD) test. **b**, Numerical error (MSE, logarithmic axis) in distribution and on held-out values: FP15 0.007 → 0.09, xVal 0.003 → 0.0035. **c**, Exact reproduction to three significant figures: FP15 99.9% → 34.5%, xVal 6.0% → 1.5%. The same evaluated predictions receive opposite rankings under approximate deviation and exact copying. Encoding, head and training change together, so this is a pipeline comparison rather than an isolated output-head effect. Central reported values from [8], Tables 12–13; source-reported $\pm$ values are retained in the figure source data.

The out-of-distribution lookup test reinforces the distinction. FP15’s three-significant-figure accuracy falls to 34.5%, while xVal records MSE 0.0035 with 1.5% exact accuracy [8].

The two cases are complementary. One representation excludes a required value; another produces small deviations that a strict consumer rejects. Neither is diagnosed by whether the final number was serialized as text. Target, support, precision, loss and delivered statistic together explain which uses the interface can support.

5. What survives a transformation?

Predictive interfaces usually transform a model’s computation before delivering a report. Mapping strings to labels, normalizing candidates, aggregating numerical scores, and enforcing constraints can improve a useful property. The improvement belongs to that property and its stated reference: a more usable report need not preserve the original model distribution, and a more faithful distribution need not produce better decisions.

An operational test is whether a requested action can still be computed from the delivered report without recovering discarded information. A distribution can supply its mean or a tail probability; its mean alone generally cannot supply the latter. This concerns the information in a report, not a guarantee that an estimated distribution outperforms a directly learned point predictor.

Table 5. Transformation properties and the conditions on their use. These are conditional mathematical properties or defined goals, not blanket guarantees for the reviewed implementations.

Transformation	Property retained or enforced	Necessary qualification
Common strictly increasing map of final scores	Ordering and argmax within a candidate set	A fixed threshold must also be transformed; differences and probability meaning need not survive
Common positive affine map of utility	Expected-utility maximizing choices	Consistent application and finite expectations; arbitrary nonlinear monotone maps do not suffice
Distribution to a chosen functional	A report for the corresponding point-loss decision	Other losses, tails, or joint events may require discarded information
Strings to semantic objects	Induced object probabilities	Aggregate mapped strings and retain failure or uncovered mass
Local valid-token masking	Restriction to reachable valid continuations	Reachability is different from complete-string conditional probability
Local scores to a constrained structure	Encoded constraints, if the solver succeeds	Feasibility does not establish correctness or a calibrated joint posterior

5.1 Mapping probabilities requires accounting for their mass

For a generated string Z and object mapping g , the induced probability of object y is

$$q_Y(y) = \sum_{z:g(z)=y} q_Z(z).$$

This pushforward operation connects answer engineering [22] to semantic uncertainty: Kuhn et al. explicitly aggregate probability within semantic equivalence classes [43], and Farquhar et al. use semantic entropy for confabulation detection [44]. Its consequences matter whenever one object has several verbalizations. In an illustrative construction, two strings express A with probabilities 0.30 and 0.25, while a single string expresses B with probability 0.45. String MAP selects B; object MAP selects A. Selecting one verbalizer per class does not, by itself, implement the semantic aggregation.

Surface Form Competition identifies the related problem of probability being divided among valid wordings, and proposes domain-conditional PMI scoring [51]. Its correction reweights candidate strings using a domain-conditioned baseline; it is not the equivalence-class summation above. Noisy-channel prompting supplies another distinct direction, evaluating the input given a candidate label [52]. These predecessors make the design options explicit: pool aliases when their event is shared, correct a candidate’s baseline when that is the identified bias, or change the scored conditional. Each operation has its own interpretation.

Semantic entropy measures uncertainty among inferred meanings. It can support failure detection, while low entropy can coexist with systematic error [44]. This is why choosing an uncertainty statistic requires both an equivalence relation and a downstream validation target.

Renormalizing selected candidates instead conditions on the selected event. It can be useful without recovering the real class posterior. Contextual calibration provides a successful frozen-model intervention on candidate scores, while explicitly distinguishing its use of “calibration” from matching confidence to correctness [9]. Numerical readouts have the same accounting issue: Section 4’s RAIL enumeration and default RAFT differ in whether selected-grid mass is renormalized. Similar point accuracy does not erase that distinction.

Failures also belong to the probability and evaluation accounting. Mapping an invalid string to a failure symbol preserves its occurrence; retrying or substituting a default defines another delivered process. LIFT’s numerical pipeline changes sampling temperature after parsing failure, permits bounded retries, and eventually substitutes the training-label mean [46]. Evaluating only valid initial outputs would assess a different system and denominator. A useful fallback should therefore be credited as part of the interface while remaining distinguishable from the model’s original prediction.

5.2 Ordering and aggregation require different numerical structure

A ranking consumer can tolerate transformations that a thresholding or aggregation consumer cannot. A common strictly increasing map preserves final-score order, but generally changes score differences and the interpretation of an unchanged threshold. If the transform is known, the threshold can be adjusted. The error is assuming that an old numerical rule retains its validation automatically.

Preference modeling supplies a concrete anchor. The Bradley–Terry form used in reward-model training is $P(a \succ b \mid x) = \sigma(r(x, a) - r(x, b))$ [53], [54]. Adding a common context-dependent constant preserves these pairwise probabilities. With a fixed logistic scale, multiplying reward differences generally changes them. This is distinct from the positive-affine invariance of expected-utility choices in Table 5. Comparisons identify relative scores within the compared candidate system on their modeled scale; they do not establish a universal zero point or arbitrary comparability across rubric dimensions.

Applying transformations before aggregation poses a separate problem. Consider nonnegative rubric vectors $A=(0,3)$ and $B=(2,2)$. Their equal-weight sums favor B, with totals 3 and 4. Squaring each coordinate preserves that coordinate’s order but reverses the summed decision: 9 versus 8. This construction explains why ordinal validity within each rubric is insufficient for arbitrary weighted aggregation. It does not allege that a reviewed reward model applies this transformation.

ArmoRM makes aggregation an empirical design choice: prompt-dependent mixing can improve the evaluated reward predictor relative to fixed mixing [40]; the controlled comparison is reported in Section 6. The result supports that aggregation, without making every rubric dimension a universally comparable utility or each mixture weight a correctness probability. Likewise, local scores used as optimization potentials need evidence for their structural consumer, even if they perform well as individual classifiers [10].

5.3 Useful summaries retain selected information

Numerical means can exploit distinctions ignored by a modal report. RAIL supplies a matched-sample decision example, and G-EVAL and TrustJudge supply scoring examples [6], [15], [31]. The gain does not imply preservation of the complete distribution. On support $\{1,2,3\}$, probabilities (0.1, 0.8, 0.1) and (0.2, 0.6, 0.2) have the same mean and unique mode, 2, but different tail probabilities and entropy. A mean report therefore cannot answer every uncertainty question. Nor is mean

universally more informative than mode: distributions can share a mean but have different modes, while others share a mode but have different means.

TrustJudge is a particularly relevant predecessor because it explicitly connects distribution-sensitive scores and pairwise decisions. In the inspected v2, one Llama-3.1-70B scoring comparison reduces score–comparison conflict from 23.32% to 14.89%. Its pairwise aggregation comparison reduces five-candidate non-transitivity from 15.22% to 4.40%, while exact match changes from 80.42% to 81.61% [31]. These are distinct module comparisons. Category breakdowns include reversals; Math is a task category in the reported Qwen2.5-7B-Instruct example, not a separate Qwen-Math checkpoint [31].

The positive findings support particular consistency improvements. The appendix’s existence argument does not prove entropy preservation, and concentrating probability can concentrate it on the wrong outcome. Bidirectional aggregation can produce swap-equivariant scores; final argmax still requires a suitable tie convention, and pairwise swap consistency does not imply transitivity over three candidates. These limits distinguish consistency, sharpness, and correctness without dismissing the measured gains.

Quantile vectors provide another useful partial report. A collection of well-estimated quantiles supports its specified quantile-loss decisions, but without assumptions about interpolation, tails, and coherence it does not uniquely determine a density [36]. Teacher-quantile coverage must also remain attached to the teacher-sampling population [12]. Neither the vector shape nor the label “predictive distribution” changes its referent.

5.4 Validity and distributional fidelity can have different consequences

Local constraints act on available next steps. Suppose two prefixes each have model probability 0.5, but their subsequent probabilities of completing a valid string are 0.1 and 1. Both admit a valid completion, so a local admissibility mask can leave their initial probabilities equal. Conditioning on eventual validity instead assigns the first prefix probability $0.05/(0.05+0.5)=1/11$. Reachability and future valid mass are different quantities.

Grammar-Aligned Decoding estimates the future-validity factor missing from ordinary local masking [11]. Its ASAp procedure improves an approximation across repeated samples; its convergence result has termination assumptions and an asymptotic limit. It does not establish exact alignment after a single sample, and its reference is the original model conditioned on a grammar rather than a distribution of correct solutions.

That distinction matters empirically. With the same model and a 2,000-sample budget per problem, the appendix reports 1,588 versus 12 correct samples for ASAp versus GCD on one synthesis problem, but 69 versus 363 on another [11]. These are comparisons between approximate samplers. They neither measure an exact conditional oracle nor separate approximation error from properties of the target distribution. They do show why semantic success cannot be inferred from the alignment objective alone.

Constrained structured prediction supplies a complementary paired result. On GENIA coreference, adding ILP constraints changes F1 from 0.721 to 0.759 for the multiple-choice scoring configuration and from 0.599 to 0.577 for verbalized confidence; measured transitivity violations disappear in both [10]. These two routes share the Mistral backbone but differ in scoring prompts and demonstration counts; each before/after pair isolates the addition of constraints.

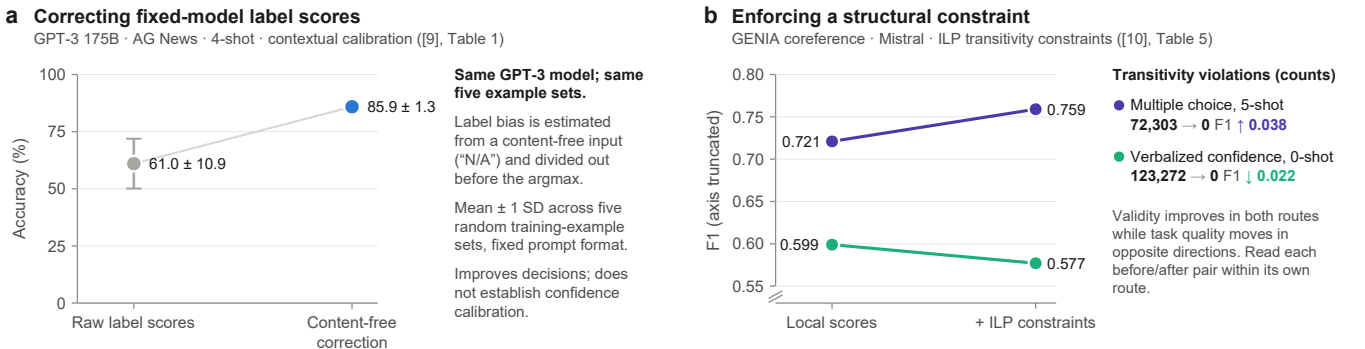


Figure 5. Transformations improve specific properties. **a**, Contextual calibration for a fixed GPT-3 175B model on four-shot AG News ([9], Table 1): estimating prompt-specific label bias from a content-free input and dividing it out before the argmax raises mean accuracy from 61.0% to 85.9% and reduces variability across prompts. Points are means and error bars are ± 1 s.d. across five random training-example sets with a fixed prompt format; the same sets are used in both conditions. This improves decisions; it does not establish confidence calibration. **b**, GENIA coreference with Mistral ([10], Table 5): adding ILP transitivity constraints to fixed local scores removes all measured violations in both scoring routes (72,303 $\rightarrow$ 0 for multiple choice with five demonstrations; 123,272 $\rightarrow$ 0 for verbalized confidence with zero demonstrations) while F1 rises in one route (0.721 $\rightarrow$ 0.759) and falls in the other (0.599 $\rightarrow$ 0.577). The F1 axis is deliberately truncated (0.53–0.80) to expose the paired changes; read each before/after pair within its own route. Model-distribution fidelity and probability calibration are not measured by either comparison.

Format-restriction experiments provide an additional, differently controlled case. Tam et al. compare natural-language prompting, format instructions, JSON mode, and a two-stage answer-then-format pipeline [55]. Their results vary by task and configuration; answer/reason field order and an LLM-based extraction step are material to interpretation. They support checking content quality and formatting jointly, not a blanket claim that all constrained decoding reduces reasoning quality. Efficient state-indexed decoding, as developed by Willard and Louf, addresses how to enforce the permitted language during generation [56]; that mechanism alone does not resolve the content-quality comparison. Evaluate structural validity, reference-law fidelity, and task quality separately.

6. What comparisons establish, and where computation is paid

The literature contains informative controls as well as heterogeneous pipeline comparisons. Their conclusions depend on what changes: a transformation of existing scores, a learned readout of fixed features, an execution policy, or the complete predictor. Computation requires the same specificity. A short delivered report can involve substantial sampling, fitting, or optimization before it reaches its consumer.

6.1 Match the attribution to the intervention

A same-model transformation comparison can isolate how existing outputs are consumed. A same-feature readout comparison studies a learned interface on frozen representations. A pipeline comparison allows representation, training, and execution to change together. These designs answer different questions rather than defining a quality hierarchy: deployment may require choosing whole pipelines, whereas a causal claim about aggregation requires a narrower intervention.

Table 6. Controls that support different levels of attribution. Metrics retain their original scales and cannot be compared across rows.

Comparison	What is held fixed	What changes and what is supported
ArmoRM: RewardBench 84.9 $\rightarrow$ 89.0 [40]	Multidimensional predictor and verbosity adjustment	Fixed versus prompt-dependent mixing; evidence for conditional aggregation in this setting
RELISH versus MLP: Pearson $\times$ 100 72.9 versus 56.3; normalized RMSE $\times$ 100 12.6 versus 17.8 [13]	Frozen backbones, label normalization/loss, approximately matched head size	Readout architecture; evidence for the trained head under these controls
CAMEL fast / always-reflect / selective: mean accuracy 76.8 / 82.9 / 81.1 [14]	Trained model and evaluation setting	Execution policy and inference work; selective execution retains part of the reflection benefit
RouteLLM Arena / Arena + GPT-4-judge labels (D_judge): MT-Bench APGR 0.580 / 0.802 [39]	Router architecture, model pair and evaluation protocol	Preference supervision; evidence about training data, not an isolated interface effect

RELISH’s aggregate covers six datasets and four backbones; dispersion across three seeds is not task heterogeneity. Its controls do not make comparisons to differently trained autoregressive systems equally isolated [13]. CAMEL’s same-model execution modes are useful, but the main-table result should not be combined with differing threshold-5 appendix entries into a single reconstructed accuracy–cost curve [14].

Evaluation targets also constrain attribution. RouteLLM’s APGR measures performance-gap recovery across strong-model call rates, not probability calibration [39]. In P(IK), the inspected Brier calculation concerns the binary event that empirical sampled success exceeds 0.5, rather than squared error against the soft success rate [38]. The report, target, denominator, and metric must be aligned before interpreting an improvement. Section 4’s teacher-coverage and exact-lookup examples show why changing that alignment can change the conclusion without changing any model output.

6.2 Reuse can occur across requests, within requests, or conditionally

Three schedules recur. **Offline supervision** transfers repeated generation into labels for a reusable head. [12] samples the teacher repeatedly, fits numerical-statistic probes, and avoids output-digit generation at deployment [12]. **Within-request reuse** retains an expensive representation while performing cheaper further computation: RELISH updates a query against fixed token memory, so three head iterations reuse one backbone pass [13]. **Conditional continuation** spends additional work on selected requests: CAMEL’s verdict margin gates reflection [14]. These schedules can coexist, but each needs its own accounting boundary.

A compact output alone establishes none of them. G-EVAL’s GPT-4 configuration obtains one scalar from 20 online ratings, and GenRM-CoT can average probabilities after 32 rationales per candidate [15], [2]. A label-producing LM-BFF demonstration configuration averages over 16 demonstration sets at deployment [4]. The consumer receives a small report while the producer executes multiple computations.

Table 7. Measured costs require a named operation and workload. These are local implementation results, not a cross-paper speed ranking.

Configuration and operation	Reported measurement	Boundary that must remain visible
[12] median probe versus numerical generation	Probe 0.034 ± 0.006 s; one sample 1.59 ± 0.08 s; 100 samples 3.28 ± 0.08 s	Single H100, averages over 100 length-20 inputs; probe includes encoding and is not the complete seven-quantile head [12]
xVal versus FP15, ERA5 training	At 500k steps: 9 h / MSE 1.75 versus 19 h / 2.14; at 70 h: MSE 1.51 versus 1.85	Four H100s; separately matched steps, tokens and runtime, with encoding/head/loss changing together [8]
I-LLMP / A-LLMP, numerical prediction and scoring	Samples: 81 / 170 s; likelihood at known targets: 1 / 3 s	Llama-2-7B, RTX3090 24GB, batch10; 10 observed and 40 queried points; loading is separately 5 s [21]

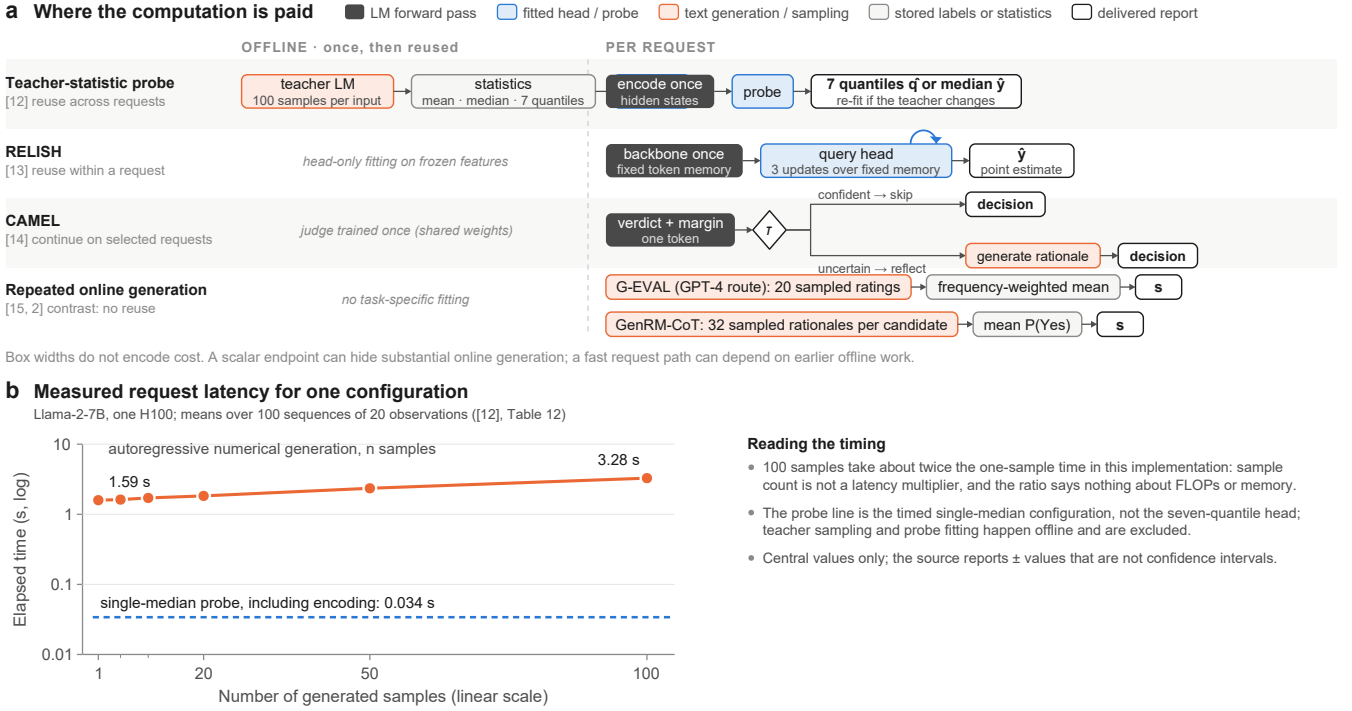


Figure 6. Compact reports come from different computation schedules. **a**, Where the computation is paid. Three schedules reuse work at different scopes: the teacher-statistic probe [12] moves teacher sampling (100 samples per training input), statistic construction and probe fitting offline and executes one encoding plus a probe per request, re-fitting if the teacher changes; RELISH [13] runs the backbone once per request and iterates a small query head three times over the fixed token memory; CAMEL [14] emits a one-token verdict with a margin and continues with reflection only when the margin falls below the threshold τ . A fourth lane shows the contrasting case of repeated online generation behind a scalar endpoint: the GPT-4 route of G-EVAL [15] aggregates 20 sampled ratings per item, and GenRM-CoT [2] averages Yes-probabilities over 32 sampled rationales per candidate. Box widths do not encode cost. **b**, Measured request latency for one configuration ([12], Table 12): generating 1/5/10/20/50/100 numerical samples takes 1.59/1.62/1.71/1.83/2.35/3.28 seconds (Llama-2-7B, one H100, means over 100 sequences of 20 observations; linear sample-count axis, logarithmic time axis), while the single-median probe takes 0.034 seconds including encoding. Sample count is therefore not a latency multiplier, and the ratio says nothing about FLOPs or memory. The probe line is the timed single-median configuration, not the seven-quantile head, and excludes offline teacher sampling and fitting. Central values only; the source reports $\pm$ values that are not confidence intervals.

The [12] measurement demonstrates why sample count is not a latency multiplier: 100 generated samples take about twice the reported one-sample time in this implementation. That observation does not establish a corresponding FLOP or memory ratio.

xVal supplies positive training-budget controls rather than a claim based solely on fewer output tokens. Its step-, token-, and runtime-matched slices support pipeline-level training efficiency, while leaving representation and head contributions entangled. NTL similarly distinguishes loss-only overhead from complete training-step overhead [47]. These examples show that careful cost attribution is already feasible and should be reused in future interface comparisons.

LLM Processes exposes another distinction: scoring a supplied numerical target and generating predictions for an unknown target are different workloads. Its likelihood timing evaluates the known target locations and values, using the numerical probability construction; it does not mean an entire arbitrary predictive distribution has been generated in one second [21]. The independent and autoregressive variants also create different dependence structures and prompt lengths. Their timings should be interpreted together with those differences, rather than as interchangeable methods returning an identical report.

Postprocessing can remain material after language-model execution. The specified 154-pair GENIA workload reports 12.42 s for multiple-choice scoring and an additional average 3.03 s for ILP [10]. Other tasks have different solver sizes and constraints. A compact score vector does not establish that its consumer is computationally negligible.

6.3 Evaluate the cost of obtaining and maintaining a report

Separate preparation, fitting, request execution, and updates. A setup-plus-recurring-cost calculation can guide a deployment only after workload-specific measurements; the reviewed studies do not supply a common break-even estimate. Batching, concurrency, and target changes belong in those measurements.

Access constraints are part of the comparison. A text-only API may support sampling but not the hidden states needed by a probe; token-logprob access may permit candidate scoring without weight access. Current author-code checks also show why scoring and sampling laws should be recorded separately, rather than inferred from shared method names. The strongest cost claim identifies what is reused, how often, under which access and workload assumptions, and with what retained consumer performance.

7. Choosing an interface and testing its reuse

7.1 Start with the consumer's requirement

For a fixed consumer, a directly delivered class or point can already contain all the information it needs. Broader reports become valuable when costs, thresholds, or uncertainty questions may change. Table 8 separates sufficient information from feasible production mechanisms; Appendix D expands the evidence, access conditions, and setup versus recurring costs. These are conditional design choices, not a cross-paper performance ranking.

Table 8. Consumer-to-interface selection guide. Sufficiency is relative to the stated use. Access and cost refer to the illustrated route; a service can expose the report while hiding its internal requirements. Setup and recurring work require workload-specific measurements.

Consumer	Information to retain	Supported path and evidence	Boundary and route-specific access	Setup / recurring cost implication
Class or candidate selection	Final label for a fixed rule; candidate scores for changed rules	Vocabulary readout or candidate head; frozen-model correction [4], [9], [45]	Label endpoint, token probabilities, or a fitted candidate model; the CoLA comparison limits a universal prompting preference [4]	Prompt scoring recurs; a fitted head adds setup
Threshold decision	Score with a validated threshold; event probability for probability-based cost rules	Correctness scoring or held-out validated margin [38], [2], [14]	Token probabilities, generated confidence, or fitted head; test calibration and discrimination separately [42], [37]	Validate threshold; gate pays continuation selectively
Ranking	Comparable order within the relevant candidate set	Known-string likelihood, rating aggregation, reward score [3], [15], [40]	Sequence probabilities, rating samples, or fitted readout; BARTScore’s criterion-specific reversal limits transfer [3]	GPT-4 G-EVAL: 20 ratings per request [15]
Weighted aggregation	Values on an aggregation-compatible scale and declared weights	Rubric heads and validated conditional mixing [40]	Hidden states plus fitted heads in ArmoRM; coordinate-wise ordering alone does not validate a weighted sum	Fit components; encode, read out and mix per request [40]
Point prediction for a loss	Mean, median, or quantile matched to target units and loss	Retained-sample statistic or corresponding point head [6], [13], [7]	Samples, grid probabilities, or hidden states; RAIL supports a functional change under fixed samples, while Huber is not automatically mean regression	Fit RELISH; one encoding plus head iterations per input [13]
Quantile or tail query	Requested quantile, validated tail probability, or adequate samples	Outcome quantile head; teacher probe when the teacher is the target [7], [12]	Adaptation weights or frozen features plus probe fitting; a median discards tails and teacher coverage concerns teacher samples	Offline samples and fitting; refresh after target changes [12]
Changing losses or joint events	Adequate distribution/sampling report; dependence for joint events	Numeric density or forecast paths [32], [18], [21]	Token probabilities or samples; distinguish known-target scoring from prediction, and marginals from joint structure	Sampling work grows with paths; scoring needs a supplied target
Valid object or coverage-oriented set	Complete object with constraint status; set with stated risk guarantee	Solver/grammar [10], [11], [17]; calibrated candidate/response sets [57], [58]	Scores plus solver, decoding control, or calibration and sampling; validity, coverage and correctness are separate criteria	Solver/decoder work recurs; coverage also needs calibration

For example, a squared-error prediction service can begin with a scalar head or a mean of available samples. If it later needs a 90th percentile, retained samples may support that request; a mean-only endpoint has discarded the necessary information. This changes the cost of adapting the consumer, while finite-sample tail accuracy remains an empirical question.

Coverage-oriented sets need precise guarantee language. Kumar et al. use split conformal prediction over fixed multiple-choice candidates, with marginal coverage under exchangeability [57]. Quach et al. use Learn Then Test to select a sampling-and-filtering procedure: with high probability over calibration data, its test risk of returning no acceptable response is bounded under the stated assumptions [58]. Acceptability is defined by the declared admission predicate; existence of an acceptable member does not validate an arbitrary member selected from the set. Set size and production cost matter alongside the guarantee. Neither type of set supplies a full density.

7.2 Fixed samples, changed consumer

Protocol A. Use RAIL’s STS-B design with 16 temperature-1 numerical samples per input as a starting point [6]. Retain samples and parsing/fallback rules. From each identical sample set compute a mean, median, empirical 90th percentile with a prespecified quantile convention, and threshold exceedance frequency. Evaluate MSE/RMSE, MAE, pinball loss at 0.9, and threshold loss against observed targets. Fit thresholds on validation data; retain failures in a common test denominator. Quantile coverage is an empirical diagnostic, not a conformal guarantee.

The first comparison changes the functional with zero resampling. A separate arm increases sample count to measure the price of accurate tails. If historical PaLM-2-S outputs or access are unavailable, declare a reproducible replacement model and treat this as a new study. The hypothesis is that retaining reports enables some consumer changes without refitting; an accuracy benefit must be measured.

7.3 Fixed scores, changed structural inference

Protocol B. Save the Mistral scores, candidates, demonstrations and document assignments for GENIA coreference [10]. Compare unconstrained decisions, a prespecified greedy repair, and the original ILP constraint set on those exact scores. Hold normalization fixed; vary it only in a separate factor.

Measure the paper’s macro-F1, explicitly specifying class/pair pooling and averaging, together with transitivity violations, solver failures, and elapsed solver time on identical hardware. Declare whether violations and their denominator count unordered triplets, directed constraints, or instantiated constraints. The paper’s 154-pair workload motivates a timing check; it is not the entire benchmark’s evaluation size. Resample documents for uncertainty, preserving within-document dependence and recomputing the same metric on each resample. Specify greedy repair order and timeout policy. The existing counterexample predicts that zero violations alone will not determine F1; both are primary outcomes.

7.4 Changed teacher, frozen versus refreshed report

Protocol C. Start with the teacher-statistic setup and its recorded constrained decoder, temperature and top-p [12]. On held-out inputs, use independent teacher samples for evaluation. First change temperature with teacher weights fixed; then test a teacher-version change separately. Compare the original frozen probe, a refitted probe, and direct sampling from the changed teacher at declared budgets. For a model-version change, declare whether the probe receives the old or updated encoder’s representations: frozen-head transfer requires compatible layers and dimensions, while an encoder change introduces representation shift as well as target shift.

Evaluate quantile loss, interval coverage against the *changed teacher’s* samples, and mean/median error against high-budget reference estimates, accounting for Monte Carlo uncertainty. Record teacher-label generation, fitting, request latency, and refresh cost separately. Where observed labels exist, evaluate them in a separate panel. The hypothesis concerns target stability and refresh costs; agreement with one teacher is not assumed to transfer to another population.

Table 9. Predeclared contrasts. These protocols are proposed studies, not new results.

Protocol	First controlled contrast	Primary outcomes	Budget-changing extension
A: consumer change	Same 16 samples, different functional	Loss-specific error, threshold loss, failures	More samples for tail estimates
B: transformation	Same scores and documents, different solver policy	F1, violations, solver failures and time	Stronger solver or scoring model in separate arms
C: refresh	Changed teacher, frozen versus refitted probe	Teacher-target fidelity and refresh cost	Label count and direct-sampling budgets

8. Limitations

This is a targeted critical synthesis. The 29 configuration-coded method studies explain mechanisms and informative controls; they are not a representative sample of all predictive uses of language models. Direct predecessors were read for the specific claims cited; detailed coverage records are available with the source ledger. Adjacent surveys support positioning, rather than increasing the empirical-evidence denominator.

The corpus contains 108 representative configurations and 53 empirical comparisons, plus two diagnostics. Extraction and independent agent checks are not blinded human double coding. No inter-rater reliability estimate, new model experiment, or full pipeline reproduction is claimed. The study matrix compresses several configurations per paper and should be read with its source records.

Version-specific PDFs anchor page-level evidence. Publication metadata may identify a journal or conference while the inspected file remains an author version; Appendix E distinguishes these. Selective code checks resolve some implementation questions, while historical-code and formula ambiguities remain disclosed, including the printed smoothing definition for price quantile regression [7]. Appendix A describes the inclusion criteria and links the underlying reading records.

The guide gives conditional choices. Studies vary in training data, scale, hardware, candidate information, and evaluation criteria, precluding a universal ranking of heads, token paths, and samples. The proposed protocols have not been run. General latent reasoning, robotics, multi-agent communication, unrestricted program synthesis, and comprehensive routing or forecasting benchmarking remain outside the scope.

The counterexamples establish that properties can separate under a particular configuration; they do not estimate how often a reversal occurs or its typical effect size. Several pivotal examples come from individual papers and tasks. Verbalized-confidence methods [41], [42], [37] illustrate an additional event-oriented score or probability report, but remain attribution sources outside the configuration-coded taxonomy.

9. Conclusion

Choose a predictive interface from the consumer's required information, then evaluate the target, transformation, and computation that produce it. The evidence supports direct reports for stable objectives, retained samples or richer summaries for specified consumer changes, and explicit validation of aggregation and structured composition. The selection guide makes these choices concrete; the configuration matrix and version-bound provenance make their support inspectable. The next empirical step is to test reuse under controlled changes of loss, constraints, and teacher process using the protocols developed here.

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Appendix A. Search, scope, and layered inclusion

A.1 Review design and evidence denominators

This is a targeted, question-driven evidence synthesis, with an audit date of 19 September 2026. It is not a preregistered systematic review, a census of language-model interfaces, or a claim of search saturation. The organizing questions concern the delivered report, its statistical target, its consumer, and the comparison needed to attribute an observed difference. A source was pursued when it supplied a direct predecessor, a distinct report mechanism, or a counterexample to a proposed inference. The search history also contains unsuccessful venue guesses and irrelevant results; these were retained rather than retrospectively presented as a predetermined protocol.

The review separates sources by their function. Counts below refer to papers unless explicitly identified as configuration or comparison records.

Table A1. Evidence layers and denominators.

Layer	Count and identifiers	Function and inclusion rule
Configuration-coded method core	29: [10], [1], [45], [39], [40], [2], [14], [13], [12], [19], [20], [16]; [4], [9], [3], [15], [11], [38], [17], [46], [32], [6], [5], [47], [8], [18]; [31]; [7], [21]	Mechanism and comparison synthesis. The existing annotation set contains 108 representative configurations, 53 empirical comparisons, and two diagnostics. A paper may contribute multiple configurations with different reports, targets, or execution paths.
Statistical theory	4: [33], [34], [35], [36]	Point forecasts, proper scoring rules, composite losses, and vector-valued elicitation. These are theoretical support, not additional empirical method comparisons.
Survey positioning	8: [24], [25], [29], [27], [28]; [22], [23], [26]	Reward models, judges, tabular prediction, confidence/calibration, and prompting. These sources establish overlap and scope; they do not enlarge the empirical-study denominator.
Targeted numeracy overlap	1: [30]	A close survey used for comparison of organizing concepts, outside the 29-study core.
Predecessor and mechanism supplement	16: [51], [52], [43], [44], [41], [42], [37], [53], [54], [57], [58], [49], [48], [55], [56], [50]	Attribution, selected mechanisms, and challenges to specific claims. Fifteen PDFs received claim-focused reading: [53] could be verified only at metadata level because its full text was unavailable. These sources do not enlarge the 108-configuration or 53-comparison count.
Excluded broad surveys	7 excluded surveys	General or adjacent surveys outside the stated scope, excluded from the bibliography and evidence counts. Their source records retain the original archive identifiers for traceability.

The 29-study scope covers finite-label decisions, reward and quality scores, real-valued predictions, and numerical predictive distributions. Structured assignments, extraction fields, and grammar constraints are included only where they expose a report-composition property. Scratch-trained numerical or time-series systems are explicitly identified as architectural bridges; they are not treated as controlled tests of the value of natural-language pretraining. Routing is one pre-answer decision example, rather than a comprehensively surveyed subfield. Broad latent reasoning, agent communication, embodied action, multimodal generation, general program synthesis, and application-specific catalogues are outside scope.

Citation following and targeted searches supplied sources for these different functions. Inclusion in the method core requires configuration-level mechanism and comparison records; relevance to positioning or attribution alone does not satisfy that criterion.

A.2 Executed searches and source handling

The documented searches contain **75 executed query records**: 42 for scope and version checks and 33 for predecessor and mechanism checks. This count denotes submitted queries, not unique search results, screened papers, or an exhaustive search history. Directly opening a publisher page, following a reference, downloading a PDF, or checking a DOI is not counted as an additional query. The logs do not provide a deduplicated search-result denominator, so no PRISMA-style retrieval or exclusion totals are inferred.

Table A2. Audited search stages.

Stage	Executed queries	Recorded scope	Resulting action
Scope and version checks	42; 16 search/access batches, 11 containing queries	Nearby surveys; numerical and probabilistic prediction; exact-title, arXiv-version, and publication checks	Included [7], [21] and the bounded [30] comparison. Rechecked metadata for 16 sources; metadata for 30 others came from existing version records without a fresh check.
Statistical and confidence predecessors	21	Surface-form competition, noisy-channel prompting, semantic uncertainty, verbalized confidence, preference modeling, and conformal prediction	Verified [51], [52], [43], [44], [41], [42], [37], [53], [54], [57], [58] identities and inspected relevant source passages where full text was available.
Numerical and structural mechanisms; routing screen	12	Numerical regression, output-format restrictions, guided decoding, forecasting ablations, and three routing candidates	Included [49], [48], [55], [56], [50] for specific mechanisms; screened three routing candidates without adding them to the supplement or the core.

Searches used web search with primary-source follow-up. The principal sources were ACL Anthology, OpenReview, arXiv, ICLR and NeurIPS proceedings, other official conference directories, and publisher pages. Publication checks also used publisher-registered metadata. The exact recorded query strings appear in A.6. A domain appearing in brackets in a query record represents the recorded search restriction, not a venue assigned to the resulting paper.

The accompanying source package records the scope and version checks in `verification/v3/search_log.json`, the [51], [52], [43], [44], [41], [42], [37], [53], [54], [57], [58] queries and access decisions in `verification/v4/predecessor_source_access.md`, and the numerical/structural queries and routing screens in `verification/v4/root_search_log.json`. Artifact identity, SHA-256 hashes, note locations, exact reading coverage, and claim-to-page links are in `verification/v4/predecessor_sources.json` and `verification/v4/root_supplement_sources.json`. Tables A3–A4 consolidate source identity and inclusion rationale; the detailed ledgers remain separate from the core configuration annotations.

Each downloaded artifact was checked against its title, authors, and inspected version before extracting page-indexed text. Its PDF hash and physical page numbers are retained in the source ledger. Publication identity and inspected content are separate: a confirmed final venue does not authorize relabeling an earlier author PDF’s pagination as the final edition. In particular, The verbalized-confidence paper [41] was read from the June 2022 arXiv v2, while its TMLR publication identity was checked separately. Source [49] likewise has a 2024 publication year and an inspected author revision dated January 2025.

When a final publication was inaccessible, an independently available public arXiv version was used with the version difference recorded. Historical attribution to [53] rests on metadata; the logistic reward expression used in the discussion was checked directly in [54]. Selected equations and a page with faulty text extraction were also rendered for visual verification. Reading did not include reproducing the methods or independently auditing every codebase.

A.3 Predecessor supplement: identity, version, and inclusion rationale

The supplement supports specific claims rather than asserting comprehensive full-paper reading or configuration coding equivalent to Appendix B. Detailed reading and non-reading ranges, including separate supplements, are recorded in the two source ledgers above. Their locators use one-based physical pages of the identified PDF.

Table A3. Direct attribution and confidence/preference predecessors.

ID and verified identity	Inspected artifact	Reason for inclusion and boundary
[51]: Holtzman et al., <i>Surface Form Competition</i> , EMNLP 2021	ACL published PDF, proceedings pp.7038–7051; 14 physical pages	Direct predecessor to the distinction between one verbalizer and an answer concept. PMI_DC reweights candidate scores by a domain prior; it does not sum semantic equivalence classes.
[52]: Min et al., <i>Noisy Channel Language Model Prompting for Few-Shot Text Classification</i> , ACL 2022	ACL published PDF, proceedings pp.5316–5330; 15 pages	Scores supplied input under candidate labels. Frozen demonstration routes and separately trained prompt/head routes must retain different comparison controls.
[43]: Kuhn, Gal and Farquhar, <i>Semantic Uncertainty</i> , ICLR 2023	arXiv:2302.09664v3, 15 April 2023; 19 pages	Direct predecessor to probability aggregation over meaning classes. A delivered entropy is a summary used for error screening, rather than automatically a correctness probability.

ID and verified identity	Inspected artifact	Reason for inclusion and boundary
[44]: Farquhar et al., <i>Detecting hallucinations in large language models using semantic entropy</i> , Nature 630,625–630 (2024)	Nature version of record, published online 19 June 2024; 12-page article-and-Methods PDF	Connects semantic variation to confabulation screening and rejection; distinguishes likelihood-based and sample-frequency estimators. Systematically wrong outputs remain a stated limitation.
[41]: Lin, Hilton and Evans, <i>Teaching Models to Express Their Uncertainty in Words</i> , TMLR, October 2022	Inspected arXiv:2205.14334v2, 13 June 2022 , not the later TMLR final PDF; 19 pages	Establishes verbalized correctness confidence as distinct from the probability of generating its tokens. Any numerical or page claim from this inspection is explicitly version-bound.
[42]: Tian et al., <i>Just Ask for Calibration</i> , EMNLP 2023	ACL published PDF, proceedings pp.5433–5442; 10 pages	Adds prompted confidence and sampled-probability routes. Calibration and selective accuracy–coverage AUC are separate metrics; this AUC is not AUROC.
[37]: Xiong et al., <i>Can LLMs Express Their Uncertainty?</i> , ICLR 2024	arXiv:2306.13063v2, 17 March 2024; 29 pages	A close predecessor already organizing prompting, sampling, and aggregation around calibration versus failure prediction. The present review inherits that consumer-specific idea and extends the evidence comparison across report roles.
[53]: Bradley and Terry, <i>Rank Analysis of Incomplete Block Designs: I. The Method of Paired Comparisons</i> , Biometrika 39(3–4),324–345 (1952)	Publisher/JSTOR metadata only; no PDF obtained	Historical attribution only. Fixed-scale logistic shift invariance discussed here is derived from the expression verified in [54].
[54]: Ouyang et al., <i>Training language models to follow instructions with human feedback</i> , NeurIPS 2022	Published 15-page main paper	Scalar reward and logistic preference likelihood provide a mechanism predecessor. This limited use does not expand the scope to a review of alignment or RLHF.

Table A4. Coverage contracts and additional mechanism checks.

ID and verified identity	Inspected artifact	Reason for inclusion and boundary
[57]: Kumar et al., <i>Conformal Prediction with Large Language Models for Multi-Choice Question Answering</i> , ICML 2023 Neural Conversational AI TEACH workshop	arXiv:2305.18404v3, 8 July 2023; 16 pages	A set-valued report for fixed multiple-choice labels. Its exchangeability-based marginal coverage does not identify a complete distribution or give arbitrary subgroup coverage. It is not an ICML main-conference paper.
[58]: Quach et al., <i>Conformal Language Modeling</i> , ICLR 2024	arXiv:2306.10193v2, 1 June 2024; 28 pages; official proceedings identity also checked	A calibrated sampling/rejection/stopping procedure. Its Learn Then Test guarantee is high probability over calibration data for test risk of failing to include an acceptable response, under the stated i.i.d. and admission-function conditions. It differs from the marginal split-conformal statement in [57].
[49]: Song et al., <i>OmniPred: Language Models as Universal Regressors</i> , TMLR, December 2024	arXiv:2402.14547v6, 30 January 2025; 24 pages	Adds the scratch-trained T5/numeric-token/sample-median route as a numerical architecture predecessor. It does not establish an isolated gain from pretrained linguistic knowledge.
[48]: Vacareanu et al., <i>From Words to Numbers: Your Large Language Model Is Secretly A Capable Regressor When Given In-Context Examples</i> , COLM 2024	arXiv:2404.07544v3, 10 September 2024, camera-ready; 55 pages	Supplies pretrained in-context regression without gradient updates, including the numeric-only instruction and default zero-temperature setting.
[55]: Tam et al., <i>Let Me Speak Freely? A Study On The Impact Of Format Restrictions On Large Language Model Performance</i> , EMNLP 2024 Industry Track	Official ACL PDF, 2024.emnlp-industry.91, proceedings pp.1218–1236; 19 pages	Tests distinct formatting strategies and task/field-order effects. It is a counterpoint to identifying valid serialization with task quality, not evidence that every constraint always harms reasoning.
[56]: Willard and Louf, <i>Efficient Guided Generation for Large Language Models</i> , arXiv preprint (2023)	arXiv:2307.09702v4, 19 August 2023; 18 pages	Adds finite-state/vocabulary indexing and guided token masking as an implementation predecessor associated with Outlines.
[50]: Tan et al., <i>Are Language Models Actually Useful for Time Series Forecasting?</i> , NeurIPS 2024	Official proceedings PDF; 30 pages	Component-removal/replacement evidence for the forecasting architectures tested. This challenges broad architectural attribution; it is not a direct ablation of every numerical language-model interface reviewed here.

The supplement connects distinct mechanisms within the stated scope. [51], [52], [43] distinguish three operations: prior-adjusted scoring, reverse conditional scoring, and aggregation over an event mapping. [41], [42], [37] distinguish the semantic

claim of a generated confidence number from both its token probability and its measured usefulness. [53], [54] distinguish fixed-scale preference likelihood from expected-utility invariance. [57], [58] expose different coverage contracts without conflating prediction sets with full outcome laws. These distinctions determine how the sources are used; their presence alone does not make the core annotation set larger.

A.4 Positioning and screened exclusions

The eight retained adjacent surveys are prompting/answer engineering [22], [23], reward-model and judge surveys [24], [25], [26], confidence/calibration surveys [27], [28], and the tabular-data survey [29]. They establish prior organizing concepts and prevent claims of exclusive novelty. Their function is distinct from the method evidence summarized in Appendix B.

[30], *Numeracy in Large Language Models: Fundamental Limitations and Paths to Improvement*, was inspected as arXiv:2608.13129v1, 13 August 2026. Its taxonomy, pretrained compatibility, training/inference stages, and computation budgets overlap with several organizing axes used here. This comparison is limited to the passages specified in `notes/v3/001_numeracy_overlap.md`; empirical numbers and general theoretical assertions from that survey are not used as method evidence.

Three routing candidates were screened to assess the boundary of the survey. Their scope is broader than the use of routing as one pre-answer decision example here. Screening did not include full-paper reading.

Table A5. Routing candidates screened without inclusion.

Candidate and original source	Actual screening depth	Decision
FrugalGPT	Official abstract	Not added. Cascading/routing remains outside a comprehensive subfield review.
Hybrid LLM	Official indexed PDF first-page text	Not added under the same bounded-scope decision; not recorded as full text read.
RouterBench	Official abstract	Not added. An evaluation-benchmark catalogue for routing is outside the interface-focused evidence synthesis.

Additional candidates received limited screening. *From Passive Metric to Active Signal* and *The Silent Vote* were inspected at official abstract level to check nearby organizing ideas. *Representing Numbers in NLP: a Survey and a Vision* was screened as a historical representation survey, and *Language Models as Interfaces, Not Oracles* as an application-specific boundary case. These sources are outside the coded core. The screening records support bounded inclusion decisions, not assertions about uninspected full papers.

A.5 Limits of this procedure

Targeted discovery can miss relevant studies, and a claim-focused supplement cannot establish exhaustive coverage of confidence elicitation, conformal inference, reward modeling, routing, or forecasting. Search-engine results depend on indexing and time. Citation following and source access were uneven, with final editions unavailable in several cases. The source records therefore preserve unresolved details and the exact inspected artifact instead of treating a publication label as a content audit.

The annotation unit is a configuration and its comparison, not an exclusive paper-level class. Epistemic statuses separate reported facts, derivations, unclear descriptions, and unchecked fields. Independent mathematical and source checks improve traceability but are not a blinded human annotation study or an experimental reproduction. Sources in the supplement may be promoted into the coded core only after the same reading, configuration, and comparison requirements are met; citing them in a targeted paragraph does not satisfy that condition.

A.6 Recorded query strings

The following lists reproduce the executed records for the three query groups in Table A2. They include refinement and identity/version queries, including unsuccessful venue guesses. They should not be interpreted as a preregistered search strategy. Direct opens and download attempts are stored with their source-access records and excluded from the query count.

A.6.1 Scope and version checks: 42 queries

The log contains 16 search/access batches, of which 11 contain query strings; the other five record direct opens or related retrieval operations.

1. survey "language models" "regression" "2025" "2026" (domain filter: arxiv.org, aclanthology.org, openreview.net)
2. survey "language models" "predictive" "interface" (domain filter: arxiv.org, aclanthology.org, openreview.net)
3. "answer engineering" survey 2025 2026 (domain filter: arxiv.org, aclanthology.org, openreview.net)
4. "language models" "numeric" "output" survey 2026 (domain filter: arxiv.org, aclanthology.org, openreview.net)
5. "large language models" regression survey (domain filter: arxiv.org, aclanthology.org, openreview.net)
6. "language models" "probabilistic regression" (domain filter: arxiv.org, aclanthology.org, openreview.net)
7. "numerical prediction" "survey" language models (domain filter: arxiv.org, aclanthology.org, openreview.net)
8. "language models" "decisions" "scores" "survey" 2025 2026 (domain filter: arxiv.org, aclanthology.org, openreview.net)
9. "Pre-train, Prompt, and Predict" 2023 2025 (domain filter: arxiv.org, dl.acm.org)
10. "The Prompt Report" "2406.06608" (domain filter: arxiv.org, openreview.net, dl.acm.org)
11. "A Survey on LLM-as-a-Judge" 2411.15594 (domain filter: arxiv.org, dl.acm.org)
12. "TrustJudge" ICLR 2026 (domain filter: openreview.net, proceedings.iclr.cc, arxiv.org)
13. "Pre-train, Prompt, and Predict" "3560815" (domain filter: dl.acm.org, acm.org, cs.cmu.edu)
14. "Making and Evaluating Point Forecasts" Gneiting (domain filter: tandfonline.com, arxiv.org, projecteuclid.org, stat.washington.edu)
15. "The Prompt Report" Schulhoff publication (domain filter: dl.acm.org, aclanthology.org, openreview.net, arxiv.org)
16. "A Survey on LLM-as-a-Judge" ACM (domain filter: dl.acm.org, arxiv.org, awesome-llm-as-a-judge.github.io)
17. "numeracy" "survey" "2025" "language models" (domain filter: arxiv.org, aclanthology.org, openreview.net)
18. "numerical" "survey" "2026" "language models" (domain filter: arxiv.org, aclanthology.org, openreview.net)
19. "Semantic Softmax" language models (domain filter: arxiv.org, aclanthology.org, openreview.net)
20. "TrustJudge: Inconsistencies" proceedings (domain filter: proceedings.iclr.cc, openreview.net)
21. "LLM Processes: Numerical Predictive Distributions" publication (domain filter: openreview.net, arxiv.org, proceedings.mlr.press, proceedings.neurips.cc)
22. "Quantile Regression with Large Language Models for Price Prediction" (domain filter: aclanthology.org, arxiv.org)
23. "From Passive Metric to Active Signal" uncertainty (domain filter: aclanthology.org, arxiv.org)
24. "The Silent Vote" "2026.gem" (domain filter: aclanthology.org)
25. "It's All in The [MASK]" (domain filter: openreview.net, jmlr.org, aclanthology.org)
26. "GLiClass: Generalist Lightweight Model" (domain filter: openreview.net, aclanthology.org)
27. "RouteLLM: Learning to Route" ICLR (domain filter: openreview.net, proceedings.iclr.cc)
28. "Generative Verifiers: Reward Modeling" ICLR (domain filter: openreview.net, proceedings.iclr.cc)
29. "CAMEL" "Adaptive" "2602.20670" (domain filter: icml.cc, proceedings.mlr.press, arxiv.org)
30. "RELISH" "regression" "2026" (domain filter: openreview.net, arxiv.org)
31. "Eliciting Numerical Predictive Distributions" ICLR (domain filter: proceedings.iclr.cc, openreview.net, arxiv.org)

32. "Generative or Discriminative" "Revisiting Text Classification" (domain filter: aclanthology.org, arxiv.org)
33. "It's All in The" "Simple Instruction-Tuning" TMLR (domain filter: openreview.net)
34. "Decoding-based Regression" TMLR (domain filter: openreview.net)
35. "CAMEL" "2602.20670" ICML (domain filter: icml.cc, arxiv.org)
36. "LLM REgression with a Latent Iterative State Head" COLM (domain filter: openreview.net, colmweb.org, arxiv.org)
37. "Simple Instruction-Tuning Enables BERT-like" "TMLR" (domain filter: openreview.net, arxiv.org)
38. "CAMEL: Confidence-Aware" 2026 (domain filter: arxiv.org, icml.cc)
39. "Numeracy in Large Language Models" "2026" (domain filter: arxiv.org)
40. "It is all in the" "100150" (domain filter: sciencedirect.com)
41. "CAMEL: Confidence-Gated Reflection for Reward Modeling" (domain filter: icml.cc, openreview.net)
42. "RELISH" "Latent Iterative" COLM2026 (domain filter: openreview.net, arxiv.org)

A.6.2 Statistical and confidence predecessor checks [51], [52], [43], [44], [41], [42], [37], [53], [54], [57], [58]: 21 queries

43. Holtzman Surface Form Competition 2021 language models
44. Min 2022 noisy channel language model prompting
45. Kuhn 2023 semantic uncertainty linguistic invariances
46. Farquhar 2024 Nature detecting hallucinations semantic entropy
47. site.aclanthology.org Surface Form Competition Holtzman
48. site.openreview.net Semantic Uncertainty Linguistic Invariances Kuhn
49. site.nature.com s41586 semantic entropy Farquhar 2024
50. site.openreview.net Teaching Models to Express Their Uncertainty in Words
51. site.aclanthology.org Just Ask for Calibration Tian 2023
52. site.openreview.net Xiong Can LLMs Express Their Uncertainty 2024
53. Kumar 2023 conformal prediction language models conformal
54. Quach 2024 Conformal Language Modeling ICLR
55. site.openreview.net "Teaching Models to Express Their Uncertainty in Words" TMLR
56. Bradley Terry 1952 Rank Analysis Incomplete Block Designs paired comparisons pdf
57. site.proceedings.neurips.cc 2022 Training language models to follow instructions human feedback Ouyang
58. site.openreview.net "Semantic Uncertainty" "VD-AYtP0dve"
59. "Teaching Models to Express" "openreview.net/forum"
60. "Bradley" "Terry" "1952" filetype:pdf site:edu
61. "Can LLMs Express Their Uncertainty" arxiv
62. "Bradley" "Terry" "1952" "pdf" "324"
63. site.openreview.net "8s8K2UZGTZ" "2022"

A.6.3 Numerical and structural mechanism checks [49], [48], [55], [56], [50] checks and routing screen: 12 queries

64. OmniPred 2024 language models universal regressors Song arxiv
65. From Words to Numbers Your Large Language Model Is Secretly A Capable Regressor COLM 2024
66. Let Me Speak Freely How the LLM learns structured output constraints EMNLP 2024 Tam
67. Are Language Models Actually Useful for Time Series Forecasting NeurIPS 2024 Tan
68. "OmniPred: Language Models as Universal Regressors" [arxiv.org, openreview.net]
69. "From Words to Numbers" "Regressor" [openreview.net, arxiv.org]
70. "Let Me Speak Freely?" [aclanthology.org, arxiv.org]
71. "Are Language Models Actually Useful" [proceedings.neurips.cc, openreview.net, arxiv.org]

72. Willard Louf 2023 Efficient Guided Generation for Large Language Models Outlines [arxiv.org]
73. "FrugalGPT" "2023" [arxiv.org]
74. "Hybrid LLM" "ICLR 2024" [openreview.net]
75. "RouterBench" Hu 2024 [arxiv.org, openreview.net]

In A.6.3, bracketed domain lists reproduce the domain filters recorded in the source log; they are not additional queries. Queries 73–75 are the three routing screens.

Appendix B. Configuration-aware method matrix

This matrix covers the **29 method studies** in the coded evidence set. Adjacent surveys used for positioning and the four statistical-theory sources have separate functions. Each study entry summarizes selected configurations rather than assigning a permanent architecture, probability meaning, or comparison grade to a paper. The annotation set contains 108 configurations and 53 empirical-comparison records, plus two diagnostic records. Configuration and comparison identifiers below resolve to that set; they are provenance identifiers, not additional bibliography entries.

Figure B1 summarizes the matrix before the study-level detail: for each study it marks the taxonomy leaf codes of the roles it realizes, so that overlap between branches of Figure 2 is visible study by study, and the training provenance of its coded configurations, which is the fitting requirement behind the access entries of Tables B.1–B.3.

Roles and training provenance of the 29 configuration-coded studies

	DELIVERED-REPORT ROLE (taxonomy leaf code)					TRAINING PROVENANCE OF THE CODED CONFIGURATIONS			
	D	S	V	P	C	Frozen model, no task fitting	Frozen features + fitted head / router	Adapted pretrained model (fine-tuning, LoRA or RL)	Trained from scratch*
LM-BFF [4]	D1		V3					●	
Contextual calibration [9]	D1					●			
[MASK] classifier [1]	D1							●	
GLiClass [45]	D2							●	
Generative vs. discriminative* [19]	D2								●
RouteLLM [39]	D3						●		
CAMEL [14]	D3							●	
BARTScore [3]		S1						●	
P(True) / P(IK) [38]		S2				●		●	
GenRM [2]		S2						●	
ArmoRM [40]		S3					●		
G-EVAL [15]		S4				●			
TrustJudge [31]		S4				●			
LIFT [46]			V1					●	
NLT [47]			V1					●	
Decoding-based Regression* [20]			V1	P3					●
RELISH [13]			V2				●		
xVal* [8]			V2						●
RAFT [5]			V3					●	
RAIL [6]			V4			●			
Teacher-statistic probes [12]			V4	P1			●		
Text-to-price quantile regression [7]			V4	P2			●	●	
LLMTime [32]				P3		●			
Chronos* [18]				P3				●	●
LLM Processes [21]				P4		●			
Prompt-based structured prediction [10]					C1	●	●	●	
GLINER2 [16]					C1			●	
Grammar-constrained decoding [17]					C2	●			
Grammar-aligned decoding [11]					C2	●			

Taxonomy leaf codes

D1 Vocabulary label readouts

D2 Class and candidate scoring

D3 Routing and selective judgment

S1 Supplied-text likelihood

S2 Event-targeted scoring

S3 Rubric readouts and mixtures

S4 Rating-probability aggregation

V1 Decoded numerical values

V2 Direct numerical heads

V3 Probability-parameterized points

V4 Extracted or learned statistics

P1 Teacher-process quantiles

P2 Observed-outcome quantiles

P3 Decoded numerical distributions

P4 Query-wise and joint predictions

C1 Scores / fields into structured objects

C2 Grammar constraints on decoded outputs

A study marked in two role columns contributes configurations to two taxonomy leaves ([4], [20], [12], [7]); a study marked in more than one provenance column has coded configurations of more than one kind (for example a frozen prompting route and a separately fine-tuned checkpoint, or a from-scratch main model and a language-model-initialized ablation). * architectural bridges whose main models train from scratch. Provenance categories follow the trainable-component entries of the configuration records summarized in Tables B.1–B.3.

Figure B1. Roles and training provenance of the 29 configuration-coded studies. Rows are the configuration-coded method studies, grouped by the branch under which they first appear in Figure 2. Left block: the taxonomy leaf code of each delivered-report role a study realizes; the codes number the 17 mechanisms of Figure 2 in reading order within each branch (three under D, four each under S, V and P, two under C) and are spelled out in the legend. A study marked in two role columns contributes configurations to two leaves ([4], [20], [12], [7]). Right block: the training provenance of the coded configurations, read from the trainable-component entries of the configuration records summarized in Tables B.1–B.3: a frozen model with no task fitting; frozen features with a fitted head or router; an adapted pretrained model (fine-tuning, LoRA or RL); or a model trained from scratch. A study marked in more than one provenance column has coded configurations of more than one kind, for example a frozen prompting route beside a fine-tuned variant ([38], [10]), a frozen-embedding comparator beside an adapted model ([7]), or a from-scratch main model beside a language-model-initialized ablation ([18]). Asterisks mark the architectural bridges whose main models train from scratch ([19], [20], [8], [18]). The figure records what the coded configurations required; it neither ranks studies nor implies that a mechanism needs a particular provenance.

Roles. D: decision; S: score; V: point-valued property; P: distribution or explicitly delimited distribution summary; C: structured composition. Arrows mark an exposed intermediate followed by its consumer. Internal probabilities count as P only when their referent is stated. In particular, RAFT’s numerical weights parameterize a point prediction, and ArmoRM’s gate weights combine rubric predictions.

Access. G means generation/sampling; L means the relevant token probabilities or logits, including scoring supplied strings; H means the relevant hidden representations; U means control of the decoding update; W means parameter adaptation. Access describes what is sufficient to implement the **specified path**, separating request-time computation from fitting. A hosted endpoint may encapsulate these operations. These are operational requirements of reviewed implementations, not lower bounds applying to every possible way of obtaining the same report. G and L are distinct API capabilities; neither is automatically implied by the other.

Controls. Transformation compares mappings of an otherwise fixed predictor; readout compares trained consumers of fixed representations; execution changes when computation is run; training changes supervision or objectives; pipeline changes several components. These descriptors apply to the named comparison, and can coexist. They identify the supported attribution, rather than rank papers by quality.

B.1 Decisions, scoring, and reward

Field	Configuration-level evidence
LM-BFF [4]	
Configurations	E04_manual_class, E04_stsb_prompt
Report / target	Report. Selected-label P to D; alternatively V for STS-B Target / supervision. Class annotations; continuous similarity targets
Production / readout	Numerical structure. Candidate normalization; two-endpoint interpolation for regression Readout. Existing masked-token vocabulary head
Access / fitting	L at masked position; W for task fine-tuning
Comparison evidence	E04_stsb_head_parameterization: backbone and few-shot protocol retained; template, head and objective change together. E04_sst2_demo also changes training context and 16-set inference ensemble.
Contextual calibration [9]	
Configurations	E05_raw, E05_contextual
Report / target	Report. Candidate-token P to D Target / supervision. Prompt-dependent label bias; class accuracy is the evaluated outcome
Production / readout	Numerical structure. Content-free bias estimate and score rescaling Readout. Existing candidate probabilities and affine correction
Access / fitting	L for task and content-free prompts; no backbone update
Comparison evidence	E05_rescaling: transformation of the same GPT-3 model and example sets; extra content-free evaluations are part of the procedure.
BARTScore [3]	
Configurations	E06_mt_cnnp, E06_mt_prompt
Report / target	Report. S for supplied text Target / supervision. Generation likelihood supplies a surrogate; evaluation uses human text-quality judgments
Production / readout	Numerical structure. Length handling, conditional log-probability averaging and prompt aggregation Readout. Teacher-forced scoring of a known target string
Access / fitting	L; W for the separately trained CNN/Para checkpoints
Comparison evidence	E06_prompt: checkpoint and scoring direction retained; conditioning phrase changes. Comparisons across checkpoints also change training.
G-EVAL [15]	
Configurations	E07_gpt4_samples, E07_gpt35_logprob
Report / target	Report. Rating-law summary to S Target / supervision. Ratings elicited by a rubric; evaluated against human judgments
Production / readout	Numerical structure. Numerical rating scale and weighted mean Readout. GPT-4 sampled ratings or text-davinci-003 rating probabilities
Access / fitting	G for the 20-sample GPT-4 route; L for the probability route; no task fitting
Comparison evidence	E07_probs: same judge/task/rubric, but discrete report, sampling policy and aggregation change; a score-resolution comparison with a documented tie effect.

Continued on next page

Method matrix (continued)

Field	Configuration-level evidence
P(True)/P(IK) [38]	
Configurations	E09_pttrue_single, E09_pik_trivia
Report / target	Report. Event-oriented P/S, optionally D Target / supervision. P(True): correctness of a supplied answer; P(IK): success of a specified answering process, estimated with 30 answers/question
Production / readout	Numerical structure. True-label probability or success-rate regression Readout. Vocabulary probability; alternatively jointly trained LM/value head
Access / fitting	L for P(True); H plus fitted value head for P(IK); G creates P(IK) targets and W fits model/head
Comparison evidence	E09_training_shift: value-head route and architecture retained; training domains change, with joint backbone/head fitting.
Single-mask instruction classification [1]	
Configurations	P14.atp100, P14.atp80_mlm20
Report / target	Report. D Target / supervision. Instruction-answer labels plus specified auxiliary objectives
Production / readout	Numerical structure. Candidate/verbalizer identities; one answer position Readout. Existing ModernBERT MLM vocabulary head
Access / fitting	L at the masked answer; W for instruction training
Comparison evidence	P14.table1.mixtures: same one-mask readout and base family; training-objective mixture and target masking change.
GLiClass [45]	
Configurations	P15.large_v3.single.zero, P15.uni_encoder.multilabel_spec
Report / target	Report. S to single-/multi-label D Target / supervision. Classification labels and staged synthetic/task supervision
Production / readout	Numerical structure. Joint text/candidate representations and similarity or learned scoring Readout. Pooled candidate/text representations and scorer
Access / fitting	Joint encoder execution/H plus scorer; W for recipe or task adaptation
Comparison evidence	P15.table5.task_adaptation: same starting checkpoint, extra labeled examples and updates. Exact large-v3 scorer/pooling combination is not pinned down in reviewed text.
RouteLLM [39]	
Configurations	P16.mf.arena, P16.mf.augmented
Report / target	Report. Preference-oriented S/P to route D Target / supervision. Arena preferences; augmentation with GPT-4-judge labels
Production / readout	Numerical structure. Learned query/model interaction and operating threshold Readout. Query embedding and matrix-factorization router
Access / fitting	Query embeddings plus fitted router; router fitting uses preference data; selected LM is called afterward
Comparison evidence	P16.table1.augmentation: same architecture/model pair/evaluation, changed supervision source. APGR measures performance-gap recovery over call rates.
ArmoRM [40]	
Configurations	P17.fixed_weights.formal2024, P17.prompt_gate.formal2024
Report / target	Report. Rubric S to aggregate S, then pairwise D Target / supervision. Named rating dimensions; separate preference-pair supervision for aggregation
Production / readout	Numerical structure. Rubric regression, verbosity adjustment, prompt-dependent weighted sum Readout. Linear 19-dimensional head on an already reward-trained backbone; prompt gate
Access / fitting	H plus fitted readout/gate; added components fitted while the reward backbone is frozen
Comparison evidence	P17-table2-fixed-vs-gated: multidimensional predictor and adjustment retained; prompt dependence and aggregator capacity change. Appendix C preserves the full record.

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Method matrix (continued)

Field	Configuration-level evidence
GenRM [2]	
Configurations	P18.gemma7b.gsm.direct, P18.gemma7b.gsm.cot32
Report / target	Report. Correctness-oriented token P/S to candidate selection Target / supervision. Correctness labels; verification-rationale supervision in CoT configuration
Production / readout	Numerical structure. Yes-token probability; soft averaging after sampled rationales Readout. Vocabulary head, direct or after rationale generation
Access / fitting	L direct; G+L for 32-rationale aggregation; W for verifier fitting
Comparison evidence	P18.direct_vs_cot32: backbone family and evaluation generator retained; supervision and request-time computation both change.
CAMEL [14]	
Configurations	P19.qwen14b.fast, P19.qwen14b.always, P19.qwen14b.gate5
Report / target	Report. Pairwise D Target / supervision. Preference judgments and reflection training
Production / readout	Numerical structure. Verdict log-probability margin gates optional reflection Readout. Initial verdict, optional text reflection, final verdict
Access / fitting	L for gate and G for reflection; W for shared judge training
Comparison evidence	P19.table1.modes: same checkpoint/evaluation pairs, different execution policy. Main and appendix threshold-5 results are retained separately.
Generative/discriminative classifier study [19]	
Configurations	P22.large_scratch.enc, P22.large_scratch.ar
Report / target	Report. Class scores/P to D Target / supervision. Supervised classes
Production / readout	Numerical structure. Class head or class-conditioned input likelihood Readout. Encoder class head versus scoring supplied input under candidate labels
Access / fitting	H/head for encoder or L for likelihood route; W for separately trained families
Comparison evidence	P22.table5.large_latency: hardware/workload/nominal size matched, inference family changes. P22.sst5.large_calibration also changes architecture, objective and training/checkpoint rules. Scratch-trained bridge.
TrustJudge [31]	
Configurations	E22.distribution_sensitive.v2, E22.likelihood_aggregation.v2
Report / target	Report. Rating-distribution-sensitive S; pairwise D Target / supervision. Prompted rating and pairwise judgments; evaluated consistency and preference agreement
Production / readout	Numerical structure. Distribution-sensitive rating transformation and bidirectional likelihood aggregation Readout. Rating/candidate likelihoods and postprocessing
Access / fitting	L for these routes; G suffices for raw-rating baseline; no judge update
Comparison evidence	E22.table1.cr_llama70b, E22.table1.ntr5_llama70b, E22.table1.em_llama70b: same judge/data, transformation or pairwise protocol changes. E22.table6.qwen_math_counterexample retains a task reversal.

B.2 Numerical values and distributions

Field	Configuration-level evidence
LIFT [46]	
Configurations	<code>E11_lift_regression</code> , <code>E11_lift_classification</code>
Report / target	Report. V or D; sampled variants available Target / supervision. Observed regression/classification labels, optimized through token CE
Production / readout	Numerical structure. Number-string serialization, parsing, retry and fallback rule Readout. Generated tokens parsed as a value/label
Access / fitting	G; fine-tuning access for the selected model; no hidden-state readout
Comparison evidence	<code>E11_insurance_servo</code> : whole pipelines compared on the same tasks. <code>E11_fpt_customers</code> : GPT-J family retained, heads and trainable parameters change.
LLMTime [32]	
Configurations	<code>E12_llmtime_density</code> , <code>E12_llmtime_samples</code> , <code>E12_llmtime_median</code>
Report / target	Report. Scored P, sampled-path P, or V Target / supervision. Frozen pretraining; observed history conditions forecasts evaluated on future observations
Production / readout	Numerical structure. Scaling, digit encoding, bin/Jacobian density conversion and sample summary Readout. Numeric token likelihood or sampled strings
Access / fitting	L for density scoring; G for trajectories/median; no LM update
Comparison evidence	<code>E12_promptcast_darts</code> : some same-backbone comparisons, but prompt and preprocessing change; specialist comparisons are pipeline comparisons. <code>E12_monash_normalization_audit</code> is a diagnostic, not an empirical gain.
RAIL [6]	
Configurations	<code>E13_rail_mean</code> , <code>E13_rail_median</code> , <code>E13_rail_enumerated_mean</code>
Report / target	Report. Loss-selected V Target / supervision. Existing LM numeric output law; observed numeric labels evaluate the chosen point
Production / readout	Numerical structure. Post-generation mean/median, candidate-grid normalization and optional probability reweighting Readout. Parsed samples or enumerated candidate likelihoods
Access / fitting	G for unweighted sample mean/median; L for enumeration or probability-weighted variants; no LM fitting
Comparison evidence	<code>E13_fixed_sampling_law_functional</code> : same sampled candidates/temperature, changed functional. <code>E13_flan_enumeration_counterexample</code> : grid/support and approximation also change.
RAFT [5]	
Configurations	<code>E14_raft_fixed</code> , <code>E14_raft_normalized</code> , <code>E14_raft_regression_head</code>
Report / target	Report. V Target / supervision. Observed numerical target; internal grid mass parameterizes the fitted point
Production / readout	Numerical structure. Numerical vocabulary weights; squared error on scalar aggregate Readout. Weighted vocabulary probabilities; scalar-head comparator
Access / fitting	L for vocabulary route or H/head for comparator; W for fitting
Comparison evidence	<code>E14_normalization_ablation</code> : grid normalization changed within the ablation setting. <code>E14_wireless_and_stsb</code> and <code>E14_music_counterexample</code> : readout/loss and inference differ across trained pipelines.
Number Token Loss [47]	
Configurations	<code>E15_standard_ntl</code> , <code>E15_single_digit_ntl</code>
Report / target	Report. V after numeric decoding Target / supervision. Numerical labels serialized as tokens; CE plus NTL
Production / readout	Numerical structure. Token-level numerical-distance loss; tokenizer determines numeric units Readout. Existing language head, numeric tokens and parser
Access / fitting	G after training; training needs logits and W; scalar-head comparator uses H
Comparison evidence	<code>E15_rjokes_tokenizer_loss</code> : loss can be compared within a fixed tokenizer; across-tokenizer and head contrasts change additional factors. <code>E15_arithmetic_counterexample</code> : loss-only exact-accuracy reversal.

Continued on next page

Method matrix (continued)

Field	Configuration-level evidence
xVal [8]	
Configurations	E16_xval_k0, E16_fp15
Report / target	Report. V Target / supervision. Observed scientific/numerical targets, with mixed token/numeric objectives
Production / readout	Numerical structure. Continuous numeric input embedding and scalar output/MSE stream Readout. Number marker plus scalar MLP; FP15 token comparator
Access / fitting	Numeric embedding/head execution; W fits backbone and heads
Comparison evidence	E16_era5_training_budget: separately matched runtime/steps, with encoding/head/loss changes. E16_lookup_consumer_reversal: same evaluation examples, two different metrics on different pipelines. Scratch-trained bridge.
Chronos [18]	
Configurations	E17_chronos_scratch, E17_chronos_lm_init
Report / target	Report. Quantized future-path P; summaries V/P Target / supervision. Observed and synthetic time series, token CE
Production / readout	Numerical structure. Historical scaling, 4,094 numerical bins and inverse mapping Readout. Autoregressive categorical tokens to paths/quantiles
Access / fitting	G from the fitted forecast model; W for original training
Comparison evidence	E17_init_ablation: initialization changes within size/forecasting recipe. E17_benchmarkII: heterogeneous trained pipelines. E17_finite_support_failure is representational analysis. Main models train from scratch.
RELISH [13]	
Configurations	P20.llama8b.relish3, P20.llama8b.mlp
Report / target	Report. V Target / supervision. Observed numerical labels
Production / readout	Numerical structure. Continuous iterative readout, label normalization and Huber loss Readout. Learned query repeatedly attends to fixed token memory, then scalar head
Access / fitting	H once per input plus head execution; head-only fitting
Comparison evidence	P20.table3.mlp_vs_relish: fixed per-backbone features, splits, loss and normalization; approximately matched head size; readout changes.
Teacher-statistic probes [12]	
Configurations	P21.llama2_7b.mean_probe, P21.llama2_7b.quantile7
Report / target	Report. V or seven-quantile P summary Target / supervision. Mean, median or quantiles of 100 samples from a specified teacher decoder
Production / readout	Numerical structure. Magnitude/value parameterization and statistic-specific supervised losses Readout. Hidden-state probe
Access / fitting	H online; G offline for teacher labels plus head fitting; teacher backbone frozen
Comparison evidence	P21.table3.mean_probe_reference: teacher/input/statistic intention retained, distilled head versus online Monte Carlo and prior fitting. Real-outcome point error and teacher-sample coverage are separate endpoints.
Decoding-based Regression [20]	
Configurations	P23.normalized, P23.unnormalized
Report / target	Report. Numeric-bin P, decoded/summarized V Target / supervision. Observed numerical targets, digit CE
Production / readout	Numerical structure. Base/precision code, normalized versus unnormalized representation Readout. Task-trained encoder plus small autoregressive decoder
Access / fitting	G for values/samples; L for bin probabilities; W for joint training
Comparison evidence	P23.table1.bbcb.encodings: common architecture/search family and task/training size, changed encoding and associated tuning/aggregation. Scratch-trained bridge.

Continued on next page

Method matrix (continued)

Field	Configuration-level evidence
Price quantile regression [7]	
Configurations	N01_point, N01_quantile, N01_knn_quantile
Report / target	Report. V or 200-quantile P summary Target / supervision. Recorded prices after the paper’s filtering procedure
Production / readout	Numerical structure. Log-price target; scalar or quantile head; smoothed pinball objective as described Readout. Adapted LM hidden states to numeric head; embedding+kNN comparator
Access / fitting	H and head/adaptor execution; W for LM adaptation; embedding access for kNN route
Comparison evidence	N01_point_quantile_amazon: head/loss/statistic and training gradients change. N01_distribution_tradeoff_amazon: representation and fitting also change. Printed smoothing definition and use of appendix interpolation remain unresolved.
LLM Processes [21]	
Configurations	N02_independent, N02_autoregressive
Report / target	Report. Marginal or ordered joint P; median V Target / supervision. Observed pairs and optional descriptive text; no task weight update
Production / readout	Numerical structure. Numeric serialization, terminal-token/bin likelihood, query ordering Readout. Prompted numeric samples or constructed token likelihood
Access / fitting	G for sampled reports; L for scoring construction; no fitting
Comparison evidence	N02_independent_autoregressive_quadratic: backbone/task/observation count/order held; adding earlier generated query values changes conditioning, dependence and execution.

B.3 Bounded structured composition

Field	Configuration-level evidence
Prompt-based structured prediction [10]	
Configurations	P13.mc.genia.local, P13.mc.genia.ilp, P13.verbal10.genia.ilp
Report / target	Report. Local S/D to constrained C Target / supervision. Coreference/structured labels; prompting, calibration and fine-tuning are separate configurations
Production / readout	Numerical structure. Local score mapping and constrained joint objective Readout. Candidate/token scores, elicited confidence or samples, followed by ILP
Access / fitting	L for MC/True-False; G for elicited/sampled variants; solver after scoring; fitted variants add parameters
Comparison evidence	P13.table5.mc and P13.table5.verbal10: fixed local scoring configuration, ILP changes joint decision; violations vanish but F1 moves in opposite directions. P13.table6.frozen_calibration additionally changes learned score mapping.
GLiNER2 [16]	
Configurations	P24.classification_single, P24.ner, P24.hierarchy
Report / target	Report. D/P or span/field outputs to C Target / supervision. Schema-conditioned class, entity and field supervision
Production / readout	Numerical structure. Class/count scores, span boundaries, hierarchical occurrence indices Readout. Shared encoder, task heads, software assembly
Access / fitting	H/task heads or equivalent task endpoint; W for original fitting
Comparison evidence	P24.table4.candidate_count: same CPU/model setting, varied label count. Component NER/classification results do not evaluate complete hierarchical objects.

Continued on next page

Method matrix (continued)

Field	Configuration-level evidence
Grammar-constrained decoding [17]	
Configurations	E10_cp_free, E10_cp_idg, E10_ed_iig
Report / target	Report. Serialized C Target / supervision. Grammar/schema; examples; input-dependent candidates in selected tasks
Production / readout	Numerical structure. Valid-token mask and constrained sequence scores Readout. Autoregressive tokens constrained by parser
Access / fitting	L+U and incremental parser; no LM update
Comparison evidence	E10_valid_vs_correct: same LM/demonstrations, constraints added. E10_candidate_information: candidate set and filtering also change available information.
Grammar-aligned decoding [11]	
Configurations	E08_gcd, E08_asap
Report / target	Report. Grammar-conditional string P to C Target / supervision. Base LM string law conditioned on grammar; task correctness evaluated separately
Production / readout	Numerical structure. Estimated future grammatical mass adjusts next-token sampling Readout. Adaptive grammar-aligned token sampling
Access / fitting	L+U, parser and adaptive cache; no weight fitting
Comparison evidence	E08_correct_positive and E08_correct_negative: same LM/problem/grammar/sample count, different decoding correction; correctness improves on one problem and worsens on another.

B.4 Reading the matrix as a design resource

Three useful starting points emerge from the controlled comparisons. If an existing report already contains the needed information, first test a consumer-specific transformation, as in RAIL or contextual calibration. If hidden representations are available and repeated requests justify fitting, compare compact readouts under a matched backbone, loss and normalization, as in RELISH. If difficult requests benefit from additional reasoning, evaluate an execution rule with the same checkpoint, as in CAMEL. These are testable choices under explicit conditions; choosing among full pipelines still requires a common task, consumer and budget.

The source and identifier crosswalk accompanying this appendix records each study’s inspected version, PDF SHA256, all included configuration IDs and comparison IDs. Appendix C illustrates fact-level tracing; Appendix D turns the matrix into consumer-specific choices. A source’s absence from a particular control category means only that this matrix records a different comparison, not that the method is intrinsically incapable of that control.

Appendix C. An annotated configuration record: ArmoRM

This example shows how the unit of evidence differs from a method name. The selected configuration, `P17.prompt_gate.formal2024`, combines nineteen adjusted rubric predictions using a prompt-conditioned gate. The delivered report is a scalar reward used for pairwise preference selection. A separate configuration, `P17.fixed_weights.formal2024`, retains the multidimensional predictor and verbosity adjustment but uses fixed mixing weights. Their linked comparison tests the aggregation change.

Source identity. [40], *Interpretable Preferences via Multi-Objective Reward Modeling and Mixture-of-Experts*, Findings of EMNLP 2024, ACL Anthology 2024.findings-emnlp.620, printed pages 10582–10592. Every locator below uses the **one-based PDF page** in the 11-page official proceedings file, not the separately retained arXiv version. PDF SHA256: 78653b7d446b051e1b97c5bb4b135cff55086e077a2e7a3aa3dad976e58bcfe9.

Record policy. The tables transcribe every populated fact in the configuration and comparison record. Values, claim IDs, epistemic statuses and source locators are retained. Field names are schema keys. `reported` records a statement in the source; `derived` records the survey’s interpretation or calculation; `not_reported_in_reviewed_sections` reports a bounded search result; `not_checked` identifies a check not performed; and `not_applicable` marks an inapplicable field. A null value is interpreted through its status. An **uncoded** field has no recorded fact; it is not converted into a claim that the paper omits the information.

The companion [C_armorm_record.json](#) preserves the configuration, comparison and source manifest, including mapping pointers and hashes. The tables below keep source locators beside the claims without repeating those identifiers.

C.1 Configuration facts

A_identity_boundary

Field / claim ID	Recorded value	Epistemic status	Source locator
<code>producer / claim_01823</code>	19-dimensional reward predictor followed by verbosity adjustment	<code>reported</code>	p. 3, 4
<code>consumer / claim_01824</code>	pairwise response selection in RewardBench	<code>reported</code>	p. 4, 5
<code>delivery_boundary / claim_01825</code>	adjusted rubric scores and aggregation rule -> scalar reward -> preferred candidate	<code>reported</code>	p. 3, 4, 5

B_target_information

Field / claim ID	Recorded value	Epistemic status	Source locator
<code>outcome / claim_01826</code>	preferred response within an evaluated pair; rubric heads separately predict named rating dimensions	<code>reported</code>	p. 3, 4, 5
<code>outcome_support / claim_01831</code>	binary pairwise preference outcome for benchmark decisions	<code>derived</code>	p. 4, 5
<code>available_information / claim_01828</code>	prompt and candidate responses; prompt features provide the gated aggregation input	<code>reported</code>	p. 3, 4
<code>prediction_horizon / claim_01829</code>	null	<code>not_applicable</code>	p. 3, 4
<code>supervision_origin / claim_01827</code>	rating annotations for component heads and preference pairs for aggregation	<code>reported</code>	p. 3, 4, 10

C_report_semantics

Field / claim ID	Recorded value	Epistemic status	Source locator
<code>roles / claim_01830</code>	["S"]	<code>derived</code>	p. 3, 4, 5
<code>report_space / claim_01832</code>	real-valued weighted aggregate of adjusted rubric predictions	<code>reported</code>	p. 3, 4

Field / claim ID	Recorded value	Epistemic status	Source locator
units_precision / claim_01833	learned reward scale, not a physical measurement or correctness-event probability	derived	p. 3, 4
probability_referent / claim_01834	null	not_applicable	p. 3, 4

D_implementation

Field / claim ID	Recorded value	Epistemic status	Source locator
backbone_origin / claim_01836	pretrained language model already trained for reward modeling; frozen for the additional readout/gate stages	reported	p. 10
readout / claim_01835	linear multidimensional predictor on a frozen previously reward-trained backbone, followed by aggregation	reported	p. 3, 4, 10
transformations / claim_01837	["verbosity adjustment", "weighted rubric aggregation"]	reported	p. 3, 4
aggregation / claim_01839	prompt-conditioned gate weights	reported	p. 3, 4, 5
numeric_geometry / claim_01838	["supervised rubric regression", "weighted aggregation"]	derived	p. 3, 4
conditioning_graph / claim_01840	prompt features -> gate weights; candidate-adjusted rubric scores -> weighted sum with those weights	reported	p. 3, 4

E_learning

Field / claim ID	Recorded value	Epistemic status	Source locator
training_protocol / claim_01841	the same multidimensional predictor and verbosity adjustment are retained in the compared ablation	reported	p. 5; Table 2
aggregation_supervision / claim_01842	preference supervision	reported	p. 4, 10
training_controls / claim_01843	null	not_reported_in_reviewed_sections	p. 5, 10; Table 2 and implementation appendix

F_compute_access

Field / claim ID	Recorded value	Epistemic status	Source locator
access_requirement / claim_01846	hidden features and trainable components are used in this method	derived	p. 3, 4, 10
online_path / claim_01844	backbone features, rubric/debias computation, aggregation and pairwise selection	reported	p. 3, 4, 10
offline_path / claim_01845	prior backbone training, feature extraction, head fitting and aggregation fitting	reported	p. 3, 4, 10
timing_result / claim_01847	null	not_checked	p. 5, 10

G_consumer

Field / claim ID	Recorded value	Epistemic status	Source locator
rule / claim_01848	compare aggregate candidate rewards for preference selection	reported	p. 4, 5
requirement / claim_01849	ordering is sufficient for fixed-pair argmax; aggregation before that step requires appropriate rubric scales	derived	p. 3, 4, 5

Field / claim ID	Recorded value	Epistemic status	Source locator
allowed_transforms / claim_01850	a common strictly increasing transform of the final scalar preserves same-pair argmax	derived	p. 3, 4, 5
verified_property / claim_01851	reported improvement on the evaluated RewardBench preference aggregate	reported	p. 5; Table 2
calibration_or_coverage / claim_01852	null	not_reported_in_reviewed_sections	p. 4, 5

H_evidence

Field / claim ID	Recorded value	Epistemic status	Source locator
comparison_links / claim_01854	P17-table2-fixed-vs-gated	reported	p. 5; Table 2
evaluation_unit / claim_01853	preference decision with benchmark aggregate score	reported	p. 4, 5
limitations / claim_01855	local aggregation ablation; not a universal interface ranking or guarantee of rubric faithfulness	derived	p. 5

C.2 Linked comparison

Comparison ID: P17-table2-fixed-vs-gated. Configurations: P17.fixed_weights.formal2024 and P17.prompt_gate.formal2024. Evidence kind: empirical_comparison.

The comparison reports a RewardBench aggregate of 84.9 versus 89.0, a derived difference of 4.1 points. It supports prompt-dependent aggregation within this ablation. The architecture/capacity of the aggregator changes alongside prompt dependence, so the control is at the fixed-predictor aggregation level. The limitations field identifies **claims outside that support**, rather than asserted findings.

Field / claim ID	Recorded value	Epistemic status	Source locator
task_and_setting / claim_03067	RewardBench	reported	p. 5; Table 2
metric / claim_03068	reported benchmark aggregate on its 0–100 scale	reported	p. 5; Table 2
results / claim_03069	{“fixed_weights”: 84.9, “prompt_gate”: 89.0}	reported	p. 5; Table 2
effect_size / claim_03070	4.1	derived	p. 5; Table 2
held_fixed / claim_03071	[“multidimensional score model”, “verbosity adjustment”, “evaluation setting”]	reported	p. 5; Table 2
changed_or_unmatched / claim_03072	[“prompt dependence of aggregation”, “aggregator architecture/capacity”]	derived	p. 3, 4, 5
uncertainty / claim_03073	null	not_reported_in_reviewed_sections	p. 5; Table 2
supported_claim / claim_03074	context-dependent aggregation improves the reported aggregate within this ablation	derived	p. 5
limitations (unsupported claims) / claim_03075	the scores are calibrated probabilities, the rubric is causally faithful, or a dedicated head universally beats generation	derived	p. 3, 4, 5

C.3 Explicit annotation gaps

The following configuration fields are present in the schema but remain uncoded in this record:

A_identity_boundary.version
A_identity_boundary.configuration_resolution
B_target_information.outcome_and_supervision_origin
C_report_semantics.report_support

```
C_report_semantics.statistic
C_report_semantics.model_event
C_report_semantics.intended_event
C_report_semantics.representation
C_report_semantics.calibration_claim
C_report_semantics.summary_limits
D_implementation.model_details
D_implementation.encoding
D_implementation.normalization
D_implementation.failure_policy
D_implementation.likelihood_object
D_implementation.method_variants
D_implementation.architecture_caveat
E_learning.trainable_components
E_learning.objective
E_learning.training_data
E_learning.training_budget
E_learning.special_information
E_learning.data_and_budget
E_learning.loss_scope
F_compute_access.timing_boundary
F_compute_access.reported_execution
F_compute_access.online_path_and_access
G_consumer.information_limits
G_consumer.symmetry_scope
G_consumer.entropy_scope
H_evidence.evaluation_scope
H_evidence.uncertainty
H_evidence.calibration_evidence
H_evidence.visual_evidence
```

The linked comparison has no encoded facts for: `dataset`, `model`, `sampling_protocol`, `data_split`, `hardware`, `budget`.

These gaps can guide later annotation without altering the claims already supported. For example, training objective details are not populated in the normalized `E_learning.objective` field, although the record does encode the readout, supervision sources and stages. A reader should inspect the versioned source before filling that field. By contrast, timing is explicitly `not_checked`, and calibration/coverage is `not_reported_in_reviewed_sections`; those statuses preserve different kinds of uncertainty.

C.4 What the example contributes

The record connects four levels that a method label alone would collapse: rubric targets, a scalar reward report, a pairwise consumer, and an aggregation comparison. Keeping them separate makes the positive conclusion usable: for these fixed rubric predictions, a learned prompt-dependent aggregator improves the evaluated preference decisions. A deployment interested in calibrated correctness or different rubric weights would specify and test that additional consumer explicitly.

Appendix D. Choosing an interface from its consumer

An interface choice begins with the action the downstream component must take. Fix its available information, candidate set, loss, operating thresholds and update schedule; then retain the part of the report that supports that action. The information field below describes **theoretical sufficiency for the stated consumer**, assuming the supplied quantities are correct. It does not rank estimators under finite data or compute. A coarser learned report can be more accurate or cheaper than a richer estimated one. Conversely, retaining samples or multiple properties may support later changes in the consumer without repeating the producer computation.

There is no representation-independent smallest report: a producer can always return the final action for one fixed consumer. The practical question is which information to expose so the consumer can apply or change its own rule. Table D1 uses that boundary. G/L/H/U/W have the operational meanings in Appendix B. The evidence field points to exact comparison records, whose controls and source versions remain inspectable.

Table D1. Consumer requirements, implementable routes, and evidence-conditioned choices.

Consumer and fixed requirement: Choose one class or candidate. Fixed candidate set and tie rule.	
Information sufficient at the consumer boundary	Final D if selection is delegated; otherwise an ordering of expected utilities suffices. Under equal-cost classification, class-probability ordering is enough for argmax.
Practical production routes	Candidate vocabulary readout; joint candidate encoder; verifier score; query-only router when the answer does not yet exist.
Supporting comparison and what it holds fixed	E05_rescaling keeps the LM and example sets fixed: correcting prompt-specific score bias improves AG News decisions. P17-table2-fixed-vs-gated keeps rubric predictions fixed and improves pairwise selection by changing aggregation.
Counterexample or boundary that changes the choice	E04_stsb_head_parameterization changes template/head/objective together; its improvement is not a universal vocabulary-head advantage. The same paper’s CoLA reversal [4] (Table 3) motivates task-specific validation. Query-only routing and answer-conditioned verification use different information.
Access for the illustrated path	L for candidate probabilities; H plus a fitted scorer for representation routes. A trained endpoint can return D directly.
Consumer and fixed requirement: Apply a threshold, route, or abstain. Fixed costs or resource tradeoff.	
Information sufficient at the consumer boundary	Final binary D if the rule is delegated; otherwise a score tied to the chosen threshold. An explicit probability threshold requires a named event and supporting probability evaluation; a margin can instead be tuned directly for measured cost/quality.
Practical production routes	Validation-tuned router score; correctness/success estimate; margin-gated continuation.
Supporting comparison and what it holds fixed	P19.table1.modes compares fast, always-reflect and selective execution using the same checkpoint. P16.table1.augmentation tests supervision for a fixed router architecture/model pair.
Counterexample or boundary that changes the choice	CAMEL’s margin is an execution signal, not a demonstrated correctness probability. RouteLLM’s APGR measures gap recovery, not calibration. E09_training_shift changes P(IK) supervision and its out-of-domain behavior; its reported Brier target is the thresholded empirical-success event.
Access for the illustrated path	L for the CAMEL gate plus G when reflection runs; query embedding plus router for RouteLLM; H/value head for fitted P(IK).
Consumer and fixed requirement: Rank candidates. Fixed items and tie policy; only order matters.	
Information sufficient at the consumer boundary	An order or scalar S whose order has been validated for the desired quality. A common strictly increasing transformation of the final scalar preserves the order.
Practical production routes	Known-string likelihood score; elicited rating; verifier score; rubric aggregate.

Continued on next page

Consumer and fixed requirement: Rank candidates. Fixed items and tie policy; only order matters. (*continued*)

Supporting comparison and what it holds fixed	E06_prompt fixes the BART checkpoint and scoring direction while changing the conditioning phrase. E07_probs shows how report resolution and rank metrics interact within one judge/task/rubric.
Counterexample or boundary that changes the choice	G-EVAL's Spearman gain and Kendall decrease involve ties and changed sampling/aggregation; they are not proof of a universal conflict between ranking consumers. BARTScore's additional paraphrase training improves one evaluation but reduces Rank19 factuality accuracy [3] (Tables 3 and 5).
Access for the illustrated path	L for known-string/verifier likelihood; G for sampled ratings; H/head for rubric scores.

Consumer and fixed requirement: Combine scores or choose under changing utilities. Fixed weights, or a declared family of later weights/costs.

Information sufficient at the consumer boundary	For one fixed weighted sum, its scalar is enough; for changing weights, preserve the component vector on appropriate scales. Ordinal rankings alone do not determine weighted sums.
Practical production routes	Multi-rubric readout with learned or specified aggregation; rating distribution retained for later summaries.
Supporting comparison and what it holds fixed	P17-table2-fixed-vs-gated holds the rubric predictor and verbosity adjustment fixed: prompt-dependent mixing improves RewardBench from 84.9 to 89.0.
Counterexample or boundary that changes the choice	Gate weights express mixing, not event probabilities. A coordinatewise monotone transform can change an aggregate ordering: vectors (0,3) and (2,2) sum to 3 and 4, but squared coordinates sum to 9 and 8. This is a constructed counterexample, not an experiment.
Access for the illustrated path	H and fitted rubric/gate components; G or L if ratings are elicited. Preserve rubric identities and scale transformations alongside the values.

Consumer and fixed requirement: Predict a number under a specified loss. Target units and transformation fixed.

Information sufficient at the consumer boundary	The relevant point property: conditional mean for squared error, a median for absolute error, a specified quantile for pinball loss, subject to the usual existence conditions [33].
Practical production routes	Direct trained head; probability-weighted numerical readout; mean/median of numeric samples; enumerated numeric grid.
Supporting comparison and what it holds fixed	E13_fixed_sampling_law_functional retains backbone, sampled candidate set and temperature: sample mean improves RMSE over sampled-string argmax. P20 . table3.mlp_vs_relish compares readouts of fixed features with matched loss/normalization.
Counterexample or boundary that changes the choice	E13_flan_enumeration_counterexample shows a restricted-grid mean can lose to greedy decoding. E14_music_counterexample favors the direct regression-head baseline. E16_lookup_consumer_reversal shows MSE and exact-copying accuracy can favor different pipelines.
Access for the illustrated path	G for unweighted sample summaries; L for candidate enumeration/RAFT; H for learned direct or iterative heads, with prior fitting.

Consumer and fixed requirement: Request a quantile, interval, or tail probability. Event, level and target population fixed.

Information sufficient at the consumer boundary	The requested quantile(s) or event probability can suffice; a whole law is optional. Arbitrary later tail queries require adequate support and a declared interpolation/tail policy.
Practical production routes	Direct quantile head; teacher-statistic probe; retained forecast samples; a stated numeric density construction.
Supporting comparison and what it holds fixed	N01_distribution_tradeoff_amazon exposes distinct coverage-error and CRPSS tradeoffs on the same filtered price data. Teacher-interval coverage [12] (Table 2) validates a different target. These are pipeline evidence and target-specific coverage, respectively.

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Consumer and fixed requirement: Request a quantile, interval, or tail probability. Event, level and target population fixed. (*continued*)

Counterexample or boundary that changes the choice	The kNN comparator for price quantile regression [7] has lower calibration error but lower CRPSS than the adapted LM; neither dominates both criteria. Seven teacher quantiles concern a teacher decoder, while price quantiles concern recorded outcomes. A finite quantile vector leaves intermediate/tail probabilities underspecified.
Access for the illustrated path	H plus fitted quantile/probe head; G for sample quantiles; L for the numerical density route. Teacher probes additionally need G offline for labels.

Consumer and fixed requirement: Consume a full predictive law or a joint law. Later events/losses or dependencies matter.

Information sufficient at the consumer boundary	A specified probability law on the relevant support; for multiple variables, enough dependence information for the consumer's joint events. Marginals alone suffice only for consumers that use marginals or under an explicit independence model. Samples provide Monte Carlo access with finite-sample error.
Practical production routes	Digit-factorized numeric-bin distribution; scored density; sampled paths; independent marginals or ordered conditional prediction.
Supporting comparison and what it holds fixed	N02_independent_autoregressive_quadratic fixes backbone/task/observation count/query order, then changes conditioning/dependence; MAE and NLL assess different uses. LLTime supplies an explicit bin-width/Jacobian density construction.
Counterexample or boundary that changes the choice	E17_finite_support_failure shows how scaling can exclude a required future value. A-LLMP's fixed-order joint need not be query-permutation-consistent. LLTime/LLMP likelihood and sampling procedures have separately specified laws.
Access for the illustrated path	L for scored likelihood; G for sampling; U where numeric restrictions modify decoding. Fitting is required for scratch-trained [20]/Chronos; frozen-LM prompting is another route.

Consumer and fixed requirement: Return a structurally legal object. Grammar/schema and semantic task fixed.

Information sufficient at the consumer boundary	A completed C if assembly is delegated; otherwise component reports plus dependencies and constraints needed by the chosen assembler. Validity and semantic quality require separate checks.
Practical production routes	Local scores plus ILP; schema-conditioned heads plus software assembly; constrained token decoder; grammar-aligned sampler.
Supporting comparison and what it holds fixed	P13.table5.mc and P13.table5.verbal10 each retain local scores while adding ILP. E10_valid_vs_correct retains LM/demonstrations and adds constraints.
Counterexample or boundary that changes the choice	The two [10] configurations both remove violations, while F1 rises in one and falls in the other. E08_correct_positive/E08_correct_negative show grammar-alignment gains in distributional fidelity do not imply uniform task-correctness gains.
Access for the illustrated path	L or G for local scoring plus a solver; H/heads for schema extraction; L+U and parser for controlled decoding.

Table D2. Setup and recurring costs for the eight consumers in Table D1, in the same order.

The entries describe the illustrated implementations, not measured cost rankings. Setup includes labels, fitting, validation and reusable preprocessing beyond the starting checkpoint. Recurring cost includes every operation needed to deliver one report; work depending on the current input belongs here even when it is called preprocessing. Record the amortization horizon, batching and refresh schedule. A sample count is not a count of serial API calls or a wall-clock multiplier.

Consumer	Setup and refresh cost	Recurring cost per report
Choose one class or candidate.	For contextual calibration [9], evaluate content-free prompts for the chosen prompt/candidate setting. Learned candidate encoders or scorers require supervision and fitting; revalidate after relevant model or label-set changes.	Candidate-token readout, supplied-string scoring, or encoder/scorer execution, followed by selection. Include all candidates required by the route; reusable calibration statistics avoid repeating their acquisition for an unchanged setting.
Apply a threshold, route, or abstain.	Fit the router/value head when used; construct its labels and validate the operating threshold against the chosen costs. Refit or retune when the model pair, event, or operating distribution changes.	Query embedding and router execution, or initial verdict and margin evaluation. CAMEL [14] adds reflection only on gated requests; routing [39] also incurs the selected model call. Report the fraction receiving extra computation.
Rank candidates.	Specify the scoring direction/rubric and validate it against the intended quality. G-EVAL [15] generates evaluation steps for the criterion without task-specific weight fitting; separately trained scoring checkpoints carry their own training cost.	Score each supplied candidate. G-EVAL's GPT-4 route estimates its rating law with 20 sampled ratings per item/criterion , then aggregates; its token-probability route is a different access and compute configuration. BARTScore [3] scores the known target string.
Combine scores or choose under changing utilities.	Fit rubric predictions, adjustments and any learned gate [40]; preserve scale definitions. A later choice of explicit weights can reuse component reports, while learning a new aggregator requires supervision and fitting.	Extract candidate features, evaluate rubric heads and any gate, then aggregate. Changing weights alone need not repeat candidate encoding when the required component vector has been retained.
Predict a number under a specified loss.	RELISH [13] uses supervised feature extraction, normalization and head-only fitting on a frozen backbone. Sample summaries [6] require no new LM fitting; validate the chosen sampling/encoding procedure and point functional.	RELISH runs the frozen backbone once per input , then applies the iterative readout to fixed token memory—three query updates in the default configuration, not three backbone passes. Sample-based routes pay for numerical draws plus parsing/aggregation; grid routes pay for candidate scoring.
Request a quantile, interval, or tail probability.	Outcome-quantile heads [7] require observed-label fitting. Teacher-statistic probes [12] use 100 valid numerical samples per training input from a specified teacher decoder to construct labels, plus feature extraction and probe fitting. Invalid attempts and filtering also belong to label construction. A changed teacher/decoder requires reassessing those labels and the fitted probe.	Execute the backbone and fitted quantile/probe heads, or draw samples and compute the requested summary. The recurring teacher-probe route [12] avoids regenerating its 100-sample label set at each request. Evaluate coverage against the named teacher or real-outcome target.
Consume a full predictive law or a joint law.	Fit scratch-trained forecasting/numerical models where applicable [18], [20]. Frozen-LM routes still require a specified encoding, support, scaling and scoring/sampling procedure [32], [21].	Score requested observations or generate paths; joint conditional prediction may require sequential conditioning across query points. Include normalization, inverse transforms and finite-sample aggregation. Storage or reuse of samples has its own cost and retains Monte Carlo error.

Continued on next page

Consumer	Setup and refresh cost	Recurring cost per report
Return a structurally legal object.	Specify constraints and prepare grammar/tokenizer structures or task heads [10], [16], [17]. Fitted score mappings and heads require training; a fixed grammar may support reusable parser preparation.	Local scoring plus the per-object ILP solve [10], or task-head execution and assembly [16]. Constrained decoding adds per-token parsing/masking [17]; input-dependent grammars must be constructed for the current input. Grammar alignment [11] additionally updates a prefix cache across sampling iterations. Solver/parser work is not universally negligible.

D.1 A practical selection procedure

1. **Write the consumer contract.** State the target, information available at prediction time, candidates, units, loss and whether the consumer will change. A route selected before generation and a candidate selected after generation are different contracts.
2. **Select an accessible baseline.** With sample-only access, use a sample summary appropriate to the consumer; with suitable token probabilities, test candidate scoring or numerical enumeration; with hidden states and fitting resources, add a matched direct-readout baseline. The comparison is between feasible paths for this deployment.
3. **Test the narrowest useful intervention.** If an existing report contains the necessary information, keep the producer fixed and change only the transformation or decision rule. If the report discards needed information, expand it or retrain the producer; record that larger intervention.
4. **Evaluate both the requested property and its cost.** Examples include decision accuracy at a fixed call budget, rank correlation with its tie convention, numeric error in the actual target units, quantile coverage plus a distributional score, or structural validity plus semantic quality. Include setup/fitting, request-time execution and updates when comparing total deployment cost.

D.2 Worked choice: one regression service, two consumers

Suppose an existing LM produces numerical samples and two downstream services request a squared-error estimate and an upper 0.9 quantile. Retaining the same sample set permits a mean and an empirical quantile to be computed without new LM calls. This statement concerns report reuse; it makes no claim about the samples' calibration to real outcomes. RAIL's fixed-sampling comparison provides evidence for changing the point functional, while a quantile consumer additionally needs target-specific coverage evaluation. If request volume warrants fitting, a direct mean/quantile readout is a candidate whose online savings must be assessed together with label construction, fitting and updates. The teacher-probe study [12] provides a teacher-statistic example; the price-quantile study [7] provides an observed-outcome example. Their different targets determine which evaluation is relevant.

The distinction between sufficiency and estimation is operationally useful: preserve rich reports when consumers are expected to change, and compare direct properties when the contract is stable and their acquisition is cheaper. The matrix supports those conditional choices while leaving the empirical tradeoff to a matched evaluation.

D.3 Coverage-oriented sets at the boundary

A calibrated set is a useful additional consumer contract. Split conformal multiple-choice prediction [57] concerns marginal coverage under exchangeability over calibration and test draws. Conformal language modeling [58] uses Learn Then Test: with high probability over calibration data, the selected sampling/filtering procedure has test risk at most its specified tolerance for returning no acceptable answer. Its admission predicate defines acceptability, and its assumptions must be retained. These are different guarantees; neither is an input-conditional guarantee, a full density, or validation of an arbitrary member selected from the set. A finite-budget search may fail to identify a useful valid procedure. Report set size, risk, failures, and total sampling/calibration cost. This targeted supplement is outside the 29-study configuration-coded denominator.

Appendix E. Version-bound evidence and implementation provenance

Citation metadata describe the publication; locators identify the exact file inspected. A journal or conference update does not transfer page numbers to that publication’s PDF. All PDF page numbers below are one-based file pages unless a separate supplement is explicitly named. Printed proceedings pagination is bibliography metadata.

This inventory has **58 references, 57 inspected primary PDFs, and one separately paginated BARTScore supplement**. Bradley–Terry [53] has verified bibliographic metadata but no acquired full text; its model is connected to the directly inspected reward-model objective in [54]. Sixteen predecessor sources support specific claims through claim-focused reading or bibliographic verification; their coverage is recorded separately from the configuration-coded method studies (Appendix A).

The complete versions, SHA-256 hashes, source URLs, reading records and claim support are in [source inventory](#). Claim-to-source locators are recorded in [main_claim_locators.json](#).

E.1 Inspected versions

ID	Function	Inspected version / artifact	SHA-256 prefix
[24]	positioning survey	arXiv:2505.02686v2 [cs.CL], 12 Jun 2025	ddc6b03e3589b4e6
[25]	positioning survey	arXiv:2504.12328v1 [cs.CL], 12 Apr 2025	76bfdc9a3833e87c
[29]	positioning survey	arXiv:2402.17944v4; 21 Jun 2024	be19be4bbc19b8cc
[27]	positioning survey	Official ACL Anthology proceedings PDF, 2024.naacl-long.366; no arXiv version marker	2fd57911c3a437a2
[28]	positioning survey	arXiv:2503.15850v2; 3 Jun 2025	3403e9ccfe1395e7
[10]	configuration-coded method	EACL 2026 official proceedings PDF	799a490b62311a50
[1]	configuration-coded method	arXiv:2502.03793v2, 10 Feb 2025	ac43658f30aa6898
[45]	configuration-coded method	arXiv:2508.07662v1, 11 Aug 2025	c621417b312c0b98
[39]	configuration-coded method	arXiv:2406.18665v4; 23 Feb 2025	c9bc9c8171cab95b
[40]	configuration-coded method	Official Findings of EMNLP 2024 proceedings PDF, 2024.findings-emnlp.620; earlier arXiv v1 retained separately	78653b7d446b051e
[2]	configuration-coded method	arXiv:2408.15240v3 [cs.LG], 22 Feb 2025	0ca07ace7efe542a
[14]	configuration-coded method	arXiv:2602.20670v2; 7 May 2026	95adc1b7b4f39aee
[13]	configuration-coded method	arXiv:2604.01206v2; 10 Jul 2026	c5fca18c60d90a34
[12]	configuration-coded method	arXiv:2603.02913v1; 3 Mar 2026	eea3d8fb5d5a60a7
[19]	configuration-coded method	arXiv:2506.12181v3, 9 Jan 2026	983369467f8e7f42
[20]	configuration-coded method	arXiv:2501.19383v2, 12 Aug 2025	fca3d9cc2aad6a69
[16]	configuration-coded method	EMNLP 2025 System Demonstrations official PDF, pp. 130–140	e44905cfa7b8e81f
[22]	positioning survey	arXiv:2107.13586v1, 28 July 2021	be4e306377703c5b
[23]	positioning survey	arXiv:2406.06608v6, 26 February 2025	06d9e15bc94457f0
[26]	positioning survey	arXiv:2411.15594v6, 19 October 2025	b9aa718f833b295d
[4]	configuration-coded method	ACL Anthology 2021.acl-long.295 proceedings PDF	5ac4744b45799af1
[9]	configuration-coded method	PMLR 139 zhao21c main and official supplement	e94f3e10fd0c94a1
[3]	configuration-coded method	NeurIPS 2021 official main and supplement	0e983994fbed8fc5
[15]	configuration-coded method	ACL Anthology 2023.emnlp-main.153 official proceedings	37e137b676440da9
[11]	configuration-coded method	NeurIPS 2024 official proceedings PDF	662afd72a11103ca
[38]	configuration-coded method	arXiv:2207.05221v4, 21 Nov 2022	1d861f7732f65769
[17]	configuration-coded method	ACL Anthology 2023.emnlp-main.674 official proceedings	38372e463f8b7142
[46]	configuration-coded method	NeurIPS 2022 official main PDF and official separate supplement	ab391a8457423adb

ID	Function	Inspected version / artifact	SHA-256 prefix
[32]	configuration-coded method	NeurIPS 2023 official main and separate supplementary PDF	d95f49db48f5476e
[6]	configuration-coded method	ACL Anthology 2024.findings-emnlp.799 official PDF	032a6f44b6a6f07e
[5]	configuration-coded method	ICLR 2025 official proceedings PDF	d384e876db6a26fa
[47]	configuration-coded method	ICML 2025 PMLR 267 official PDF; SHA256 4b06b57f89d8e1ebcb35a65d2830321b8f3781ceb7279f2831f0e77edeef2eb09	4b06b57f89d8e1eb
[8]	configuration-coded method	arXiv:2310.02989v2, 15 Dec 2024	3b7de6d77ec14348
[18]	configuration-coded method	TMLR 10/2024 official PDF	28f427366349670f
[33]	statistical theory	arXiv:0912.0902v2, margin 7 March 2010; title-page date 26 November 2024, cause unverified	6b6cc2f0fd3daa1d
[34]	statistical theory	Published journal PDF, March 2007	d31a0c5f0ae8fec1
[35]	statistical theory	Published journal PDF, September 2010	a0568d66c353dbad
[36]	statistical theory	Publisher PDF, 2015; internal pages 1–18	bff4d0ecd014fb3a
[31]	configuration-coded method	arXiv:2509.21117v2, 26 September 2025	2e15d181a528aa92
[7]	configuration-coded method	Official ACL Anthology proceedings PDF, printed pages 12396–12415; no arXiv version asserted	73b326eb93c3445d
[21]	configuration-coded method	NeurIPS 2024 proceedings PDF	120442add75869f2
[30]	targeted overlap reading	arXiv:2608.13129v1, 13 Aug 2026	c7e2325cb88055de
[51]	claim-focused predecessor	ACL Anthology published version, pp.7038–7051	c32c3b626219a8d5
[52]	claim-focused predecessor	ACL Anthology published version, pp.5316–5330	bdfbf76c2b99964c
[43]	claim-focused predecessor	arXiv:2302.09664v3, 2023-04-15, marked ICLR 2023	0cc1c0b189dbe6cc
[44]	claim-focused predecessor	Nature version of record, 2024-06-19; 12-page article+Methods PDF	52e7ad2740b25cc7
[41]	claim-focused predecessor	arXiv:2205.14334v2, 13 June 2022; distinct from the October TMLR publication	8290fc142e96b940
[42]	claim-focused predecessor	ACL Anthology published version, pp.5433–5442	c1353178d651220e
[37]	claim-focused predecessor	arXiv:2306.13063v2, 2024-03-17, marked ICLR 2024	65fa9f9494f00bfb
[53]	claim-focused predecessor	Metadata only; full text unavailable	—
[54]	claim-focused predecessor	NeurIPS 2022 published main paper, 15 pages; separate supplement not downloaded	4ed5ef926ed5ad38
[57]	claim-focused predecessor	arXiv:2305.18404v3, 2023-07-08; first page identifies workshop	4eb8bf72a24fd947
[58]	claim-focused predecessor	arXiv:2306.10193v2, 2024-06-01, marked ICLR 2024; official ICLR proceedings identity checked	910cc4b8f4117846
[49]	claim-focused predecessor	arXiv:2402.14547v6, 30 Jan 2025; publication metadata TMLR 12/2024	0c83277a66def3a6
[48]	claim-focused predecessor	arXiv:2404.07544v3, 10 Sep 2024; COLM 2024 camera-ready	0e8068632d538b44
[55]	claim-focused predecessor	ACL Anthology 2024.emnlp-industry.91 official PDF; pp.1218–1236	c8145f4d94b41128
[56]	claim-focused predecessor	arXiv:2307.09702v4, 19 Aug 2023	8888aa2043588e41
[50]	claim-focused predecessor	NeurIPS 2024 official proceedings PDF, 30 pages	3a989e2458cc9846
[3] (supplement)	Separate supplement	NeurIPS 2021 supplement	3964ca7c73c73488

The [3] supplement is a different file from the main PDF. Its page 2 is never interpreted as page 2 of the main article. The source inventory records a separate full hash for the supplement.

The registered publication title for [16] is *GLiNER2: Schema-Driven Multi-Task Learning for Structured Information Extraction*. The inspected PDF header instead reads *GLiNER2: An Efficient Multi-Task Information Extraction System with Schema-Driven Interface*. The source identity and bibliography preserve that distinction.

E.2 Claim-to-source locators

Each claim summary is paired with a supporting location in the inspected file. Comparison controls remain specified in the body and annotation ledger. Citation IDs resolve through E.1 to the exact source version.

Record / section	Source	Claim anchor	Locator
M001 / §3	[4]	LM-BFF normalizes masked-position logits over selected label words and maps those words to classes [4]	PDF p.4, Eq. 1
M002 / §3	[4]	For STS-B, two endpoint words instead parameterize a similarity estimate: their weights are trained from the continuous target and interpolated into a point prediction [4]	PDF pp.4–5
M003 / §3	[1]	Its MMLU ablation changes performance substantially while retaining the readout and changing the training-objective mixture [1]	PDF p.4, Table 1
M004 / §3	[4]	Under LM-BFF’s few-shot protocol, manual prompting improves STS-B but loses to ordinary fine-tuning on CoLA: 9.3 (standard deviation 7.3) versus 33.9 (14.3) Matthews correlation on the reported percent scale [4]	PDF p.7, Table 3
M005 / §3	[9]	Contextual calibration estimates prompt-specific label biases from content-free inputs; for GPT-3 175B, four-shot AG News accuracy changes from $61.0 \pm 10.9\%$ to $85.9 \pm 1.3\%$ [9]	PDF pp.5–7, Table 1
M006 / §3	[3]	The prompt ensemble averages target-token log probabilities within each prompt and then averages over prompts [3]	supplement PDF p.2, Eq. 1
M007 / §3	[3]	Fine-tuning and prompt choice change its relationship with human quality: further paraphrase training improves the reported WMT19 correlation but reduces Rank19 factuality accuracy relative to the CNN/DailyMail model [3]	PDF pp.7–8, Tables 3 and 5
M008 / §3	[15]	On SummEval, GPT-4 probability aggregation changes mean Spearman correlation from 0.502 to 0.514, whereas Kendall correlation changes from 0.446 to 0.418 [15]	PDF p.4, Table 1
M009 / §3	[2]	GenRM-CoT adds verification-rationale supervision and averages Yes probabilities after sampled rationales, using 32 rationales per candidate by default [2]	PDF pp.4–6
M010 / §3	[2]	Direct GenRM is comparable to or slightly better than the discriminative comparator in the reported additional evaluation; larger gains involve the rationale configuration [2]	PDF p.23, Figure D.1
M011 / §3	[38]	Thirty sampled answers per training question estimate that process’s success, and the language model and a separate value head are trained jointly [38]	PDF pp.14, 26–27

Record / section	Source	Claim anchor	Locator
M012 / §3	[40]	Missing rubric labels are masked from regression training, and similar dimensions can retain different annotation sources [40]	official PDF pp.3–5, 10–11
M013 / §4	[6]	For PaLM-2-S on STS-B, selecting the highest-probability sample gives RMSE 1.019, whereas the ordinary sample mean gives 0.641 [6]	PDF p.10, Table 4
M014 / §4	[6]	Enumeration also has limits: for FLAN-T5-XL, greedy, enumerated-mean and sampled-mean RMSE are 0.713, 0.741 and 0.676 [6]	PDF p.5, Table 3
M015 / §4	[5]	Good point RMSE consequently need not identify a full outcome distribution. The normalized and unnormalized variants have similar STS-B RMSE at the displayed precision [5]	PDF p.28, Table 16
M016 / §4	[5]	For Gemma-2 2B on Amazon Wireless, RAFT reports RMSE 0.47, a regression head 0.51 and CE with greedy decoding 0.70; on STS-B, RAFT and the regression head both report 0.54 [5]	PDF p.8, Table 3
M017 / §4	[47]	On T5-small/rJokes, ordinary-tokenizer RMSE changes from 2.01 with CE to 1.97 with NTL; single-digit tokenization changes the corresponding comparison from 2.02 to 1.81. A scalar MSE head gives 1.82 [47]	PDF p.8, Table 6
M018 / §4	[21]	Its terminal-token likelihood construction also does not guarantee equality with numerical-bin mass under its sampling procedure [21]	PDF pp.3–4, 14–16, 22
M019 / §4	[7]	The paper’s printed smoothing definition contains an inconsistent coefficient; the implemented objective has not been independently verified, so we use the reported setup and results rather than derive guarantees from that formula. Its median improves Amazon MAPE from the point-head baseline’s 20.81% to 16.86% [7]	PDF p.6, Table 2
M020 / §4	[7]	On Amazon, Mistral’s quantile head has calibration error 0.042 and CRPSS 0.75; the embedding+kNN baseline has lower calibration error 0.010 but lower CRPSS 0.34 [7]	PDF p.7, Table 3
M021 / §4	[12]	Its nominal 95% interval has approximately 95.4–96.1% coverage of teacher samples in the reported synthetic settings [12]	PDF p.6, Table 2
M022 / §4	[12]	Real-next-value point error is tested separately: one mean-probe comparison reports MSE 0.0562 against 0.0555 for the sampled teacher mean [12]	PDF p.7, Table 3
M023 / §4	[18]	In the paper’s periodic construction, one unit spike every n observations gives $s = 1/n$. For $n > 15$, a future unit spike lies outside output support [18]	PDF p.19, Figure 16
M024 / §4	[8]	Its number-lookup result exposes another failure mode: MSE favors xVal over FP15, 0.003 versus 0.007, while three-significant-figure accuracy favors FP15, 99.9% versus 6% [8]	PDF p.26, Table 12

Record / section	Source	Claim anchor	Locator
M025 / §4	[8]	FP15’s three-significant-figure accuracy falls to 34.5%, while xVal records MSE 0.0035 with 1.5% exact accuracy [8]	PDF p.26, Table 13
M026 / §5	[9]	Contextual calibration provides a successful frozen-model intervention on candidate scores, while explicitly distinguishing its use of “calibration” from matching confidence to correctness [9]	PDF p.5, footnote 4
M027 / §5	[46]	LIFT’s numerical pipeline changes sampling temperature after parsing failure, permits bounded retries, and eventually substitutes the training-label mean [46]	main PDF pp.3–4
M028 / §5	[31]	Its pairwise aggregation comparison reduces five-candidate non-transitivity from 15.22% to 4.40%, while exact match changes from 80.42% to 81.61% [31]	PDF p.7, Table 1
M029 / §5	[31]	Category breakdowns include reversals; Math is a task category in the reported Qwen2.5-7B-Instruct example, not a separate Qwen-Math checkpoint [31]	PDF p.18, Table 6
M030 / §5	[11]	With the same model and a 2,000-sample budget per problem, the appendix reports 1,588 versus 12 correct samples for ASAP versus GCD on one synthesis problem, but 69 versus 363 on another [11]	PDF p.20, Table 1
M031 / §5	[10]	On GENIA coreference, adding ILP constraints changes F1 from 0.721 to 0.759 for the multiple-choice scoring configuration and from 0.599 to 0.577 for verbalized confidence; measured transitivity violations disappear in both [10]	PDF p.8, Table 5
M032 / §6	[40]	ArmoRM: RewardBench 84.9 → 89.0 [40]	PDF p.5, Table 2
M033 / §6	[13]	RELISH versus MLP: Pearson×100 72.9 versus 56.3; normalized RMSE×100 12.6 versus 17.8 [13]	PDF p.8, Table 3; p.26
M034 / §6	[14]	CAMEL fast / always-reflect / selective: mean accuracy 76.8 / 82.9 / 81.1 [14]	PDF p.6, Table 1
M035 / §6	[39]	RouteLLM Arena / Arena + GPT-4-judge labels (D_judge): MT-Bench APGR 0.580 / 0.802 [39]	PDF p.7, Table 1
M036 / §6	[14]	CAMEL’s same-model execution modes are useful, but the main-table result should not be combined with differing threshold-5 appendix entries into a single reconstructed accuracy–cost curve [14]	PDF p.20, Table 4
M037 / §6	[39]	RouteLLM’s APGR measures performance-gap recovery across strong-model call rates, not probability calibration [39]	PDF pp.4–5
M038 / §6	[38]	In P(IK), the inspected Brier calculation concerns the binary event that empirical sampled success exceeds 0.5, rather than squared error against the soft success rate [38]	PDF p.25, Appendix A.3
M039 / §6	[4]	A label-producing LM-BFF demonstration configuration averages over 16 demonstration sets at deployment [4]	PDF p.13

Record / section	Source	Claim anchor	Locator
M040 / §6	[12]	[12] median probe versus numerical generation Probe 0.034 ± 0.006 s; one sample 1.59 ± 0.08 s; 100 samples 3.28 ± 0.08 s Single H100, averages over 100 length-20 inputs; probe includes encoding and is not the complete seven-quantile head [12]	PDF p.25, Table 12
M041 / §6	[8]	xVal versus FP15, ERA5 training At 500k steps: 9 h / MSE 1.75 versus 19 h / 2.14; at 70 h: MSE 1.51 versus 1.85 Four H100s; separately matched steps, tokens and runtime, with encoding/head/loss changing together [8]	PDF p.6, Table 2
M042 / §6	[21]	I-LLMP / A-LLMP, numerical prediction and scoring Samples: 81 / 170 s; likelihood at known targets: 1 / 3 s Llama-2-7B, RTX3090 24GB, batch10; 10 observed and 40 queried points; loading is separately 5 s [21]	PDF p.20, Table F.1
M043 / §6	[47]	Its step-, token-, and runtime-matched slices support pipeline-level training efficiency, while leaving representation and head contributions entangled. NTL similarly distinguishes loss-only overhead from complete training-step overhead [47]	PDF p.6
M044 / §6	[10]	The specified 154-pair GENIA workload reports 12.42 s for multiple-choice scoring and an additional average 3.03 s for ILP [10]	PDF p.9, Table 8

E.3 Source-backed supplementary claims

The following specific claims are supported by the stated source locations. Detailed reading coverage remains in the machine-readable source ledger. [53] supplies verified bibliographic metadata; no full-text reading or PDF page locator is asserted for that source.

Source	Claim checked	Inspected-file pages / locator
[51]	Surface-form competition; domain-conditional PMI reweights candidate strings rather than summing a semantic equivalence class.	pp.1–2, 4–5, §§1–3
[52]	Noisy-channel prompting scores the known input given a candidate label; it changes the scored conditional.	pp.2–4, §2
[43]	Semantic equivalence classes aggregate answer probability before an entropy statistic is formed.	pp.4–5, Eq.2 and entropy construction
[44]	Semantic entropy targets confabulations; low semantic entropy need not detect systematic errors. Practical estimators differ from an exact full-law sum.	pp.1–2, relevant passages on p.3, Methods pp.7–8; inspected Nature PDF
[41]	Model training can make generated confidence express uncertainty about answers; the inspected preprint is separate from final journal metadata.	pp.1–6; inspected arXiv v2
[42]	Prompted verbal confidence is evaluated against correctness. The Labelprob estimate can use ten sampled labels; the reported AUC is selective accuracy–coverage, not AUROC.	pp.2–4; methods and evaluation
[37]	Calibration and failure detection call for different aggregation strategies; this is a direct consumer-aware predecessor.	p.3 Fig.1; p.6 metrics; pp.8–9 recommendations
[53]	Paired-comparison model identity and publication metadata only; full text was not acquired.	No inspected-PDF locator
[54]	The pairwise reward objective uses the logistic of reward-score differences.	p.6 Eq.1
[57]	Split conformal prediction on multiple-choice candidates provides marginal coverage under exchangeability.	pp.2, 5; construction and guarantee

Source	Claim checked	Inspected-file pages / locator
[58]	Learn Then Test controls test failure risk with high probability over calibration data; the admission predicate defines an acceptable response.	pp.2, 4–6; Eq.1 and calibration procedure
[49]	A scratch-trained 200M T5 uses custom numerical tokens and response cross-entropy; a median aggregates decoded samples.	pp.4–5
[48]	Pretrained in-context regression uses examples and numeric responses without gradient updates; prompting, precision, and prior training enter the comparison.	pp.1–6
[55]	Format instructions, JSON mode and answer-then-format pipelines have task-dependent effects; field order and answer extraction are relevant controls.	pp.2–6
[56]	Finite-state structure and vocabulary indexing support constrained token selection.	pp.1–4
[50]	Removing or replacing LM components in One Fits All, Time-LLM and CALF can match or improve forecasting; these are not direct LLTime or Chronos ablations.	pp.1–6

E.4 Figures, uncertainty notation and implementation caveats

- Figure 1 is a conceptual overview with illustrative values, not measured study results. Its [caption](#) specifies the interpretation of report roles and schematic geometry.
- Figures 3–6 use [version-bound captions](#) and [source data](#); Figure 2 is generated from `taxonomy_tree_body.tex`, with row membership recorded in `taxonomy_membership.json`; Figure B1 is drawn by `make_figures_v6.py` from a study table whose leaf codes repeat that membership and whose provenance columns follow the trainable-component entries of the configuration records. The [numerical source checks](#) document the plotted values, which are unchanged from the previous figure set. Source-reported dispersion is preserved without assigning an unverified definition.
- Figure 4 uses xVal [8], inspected arXiv:2310.02989v2, pp.25–26, Tables 12–13. Panel a illustrates the look-up task and the training range with the withheld interval; panels b and c plot the same lookup task and only the two named pipeline rows, in distribution and on held-out values, with a logarithmic MSE axis from 0.001 to 0.2 and a linear accuracy axis.
- Figure 5a uses [9], official ICML 2021 PDF pp.6–7, Table 1. The $\pm$ values are standard deviations across five training-example sets at a fixed prompt format, with the same sets before and after correction. They are not variances or confidence intervals.
- Figure 5b uses [10], inspected PDF p.8 Table 5 and p.26 Table 21. The two routes share Mistral and differ in scoring prompts and demonstrations. The configuration identifier verbal10 names a confidence scale, not ten demonstrations. The selected confidence row is zero-shot; multiple choice is five-shot. The F1 differences printed beside the violation counts (0.038 and 0.022) are differences of the reported values, not additional measurements.
- Figure 6a is schematic; its node classes (model forward pass, fitted head or probe, text generation or sampling, stored labels or statistics, delivered report) describe the illustrated implementations, and its fourth lane restates the 20-rating [15] and 32-rationale [2] configurations discussed in Section 6.2. Figure 6b uses [12], inspected PDF p.25 Table 12. Only the vertical timing axis is logarithmic; all six sample counts are labeled. The probe measurement concerns the single median, includes encoding, and excludes teacher-label creation and fitting.
- [7]’s printed smoothing definition has an inconsistent coefficient. This is a source-description ambiguity, not an independently demonstrated implementation bug. The reported setup is usable evidence, while deriving an elicitation guarantee from the printed expression is not justified.
- [7] monotone increments/interpolation, [32] density versus sampling semantics, and [21] terminal-token likelihood versus sampled-bin mass are qualified in the [implementation audit](#). The ledger records source interpretation and code inspection rather than experimental reproduction.

The normalized [108 configuration records](#) and [55 comparison/diagnostic records](#) are preserved in the source ledger. Appendix C illustrates both populated and uncoded fields. A missing annotation means it was not coded; it is not evidence that the source omitted the fact.

Appendix F. Numerical mechanism map

a Where numerical structure enters shaded cells mark the principal numerical intervention of each configuration

	ENCODING / TARGET	READOUT	TRAINING LOSS	CONSUMER FUNCTIONAL
RAIL [6] frozen LM	Existing numeric serialization	Samples or candidate mass	Unchanged	• Loss-matched mean / median
RAFT [5] adapted LM	Numerical token values	• Vocabulary-weighted sum	• MSE on the aggregate	Point prediction
NTL [47] adapted LM	• Token values; tokenizer matters	Existing token head	• CE + numerical token loss	Parsed number
RELISH [13] frozen features	Input token representations	• Iterative scalar head	• Huber	Point prediction
Price QR [7] LoRA-adapted LM	Text input; log-price target	• 200 quantiles	• Smoothed pinball (as printed)	Median / quantiles
xVal* [8] from scratch	• Continuous number embedding	• Scalar number head	• Token CE + number MSE	Numerical fields
Chronos* [18] from scratch	• Scale + quantize	• Categorical numerical bins	Cross-entropy	• Dequantize; forecast summary

* main models trained from scratch (architectural bridges). Columns organize mechanisms; they do not imply a chronological progression or a matched accuracy comparison. [12] teacher-statistic labels and [7] observed-outcome labels remain distinct targets.

b Support matters: a Chronos construction

[18], Fig. 16: representable range is $[-15s, 15s]$ with s the historical mean absolute value

one unit spike every $n = 20$ observations $\rightarrow s = 1/n = 0.05$

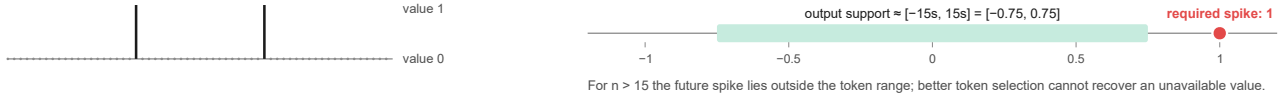


Figure F1. Where numerical structure enters. **a**, Seven numerical-report configurations placed by the stage at which their principal numerical intervention acts: encoding or target, readout, training loss, or the consumer’s functional (shaded cells). RAIL leaves training unchanged and selects a loss-matched statistic at inference [6]; RAFT changes the readout and places MSE on the aggregate [5]; NTL adds a numerical token loss and depends on the tokenizer [47]; RELISH adds an iterative scalar head trained with a Huber loss [13]; the price quantile head predicts 200 quantiles with a smoothed pinball loss as printed [7]; xVal changes encoding, head and loss together [8]; Chronos scales and quantizes the series into categorical bins and dequantizes forecasts [18]. Asterisks mark main models trained from scratch. Columns organize mechanisms and imply neither a chronological progression nor a matched accuracy comparison; Table 3 is the primary in-text comparison, and [12] teacher-statistic labels and [7] observed-outcome labels remain distinct targets. **b**, Support matters: in Chronos the representable range is $[-15s, 15s]$ with s the historical mean absolute value [18]. With one unit spike every $n = 20$ observations, $s = 0.05$ and the output support is approximately $[-0.75, 0.75]$, so the required spike of 1 lies outside the token range for any $n > 15$; better token selection cannot recover an unavailable value.

The [editable SVG](#) and [figure source data](#) provide the underlying asset. Section 4 discusses the OmniPred and in-context regression predecessors and the bounded forecasting-ablation counterpoint [49], [48], [50]; the configurations shown here are the seven mechanism rows of Table 3 that change where numerical structure enters.